Python Setup and Usage

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Guido van Rossum and the Python development team

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Python Software Foundation
Email: docs@python.org

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This part of the documentation is devoted to general information on the setup of the Python environment on different platforms, the invocation of the interpreter and things that make working with Python easier.

CHAPTER

ONE

COMMAND LINE AND ENVIRONMENT

The CPython interpreter scans the command line and the environment for various settings.

CPython implementation detail: Other implementations' command line schemes may differ. See implementations for further resources.

1.1 Command line

When invoking Python, you may specify any of these options:

```
python [-bBdEhiIOqsSuvVWx?] [-c command | -m module-name | script | - ] [args]
```

The most common use case is, of course, a simple invocation of a script:

```
python myscript.py
```

1.1.1 Interface options

The interpreter interface resembles that of the UNIX shell, but provides some additional methods of invocation:

- When called with standard input connected to a tty device, it prompts for commands and executes them until an EOF (an end-of-file character, you can produce that with `Ctrl-D` on UNIX or `Ctrl-Z, Enter` on Windows) is read.
- When called with a file name argument or with a file as standard input, it reads and executes a script from that file.
- When called with a directory name argument, it reads and executes an appropriately named script from that directory.
- When called with `-c command`, it executes the Python statement(s) given as *command*. Here *command* may contain multiple statements separated by newlines. Leading whitespace is significant in Python statements!
- When called with `-m module-name`, the given module is located on the Python module path and executed as a script.

In non-interactive mode, the entire input is parsed before it is executed.

An interface option terminates the list of options consumed by the interpreter, all consecutive arguments will end up in `sys.argv` – note that the first element, subscript zero (`sys.argv[0]`), is a string reflecting the program's source.

-c <command>

Execute the Python code in *command*. *command* can be one or more statements separated by newlines, with significant leading whitespace as in normal module code.

If this option is given, the first element of `sys.argv` will be `"-c"` and the current directory will be added to the start of `sys.path` (allowing modules in that directory to be imported as top level modules).

Raises an auditing event `cpython.run_command` with argument `command`.

-m <module-name>

Search `sys.path` for the named module and execute its contents as the `__main__` module.

Since the argument is a *module* name, you must not give a file extension (`.py`). The module name should be a valid absolute Python module name, but the implementation may not always enforce this (e.g. it may allow you to use a name that includes a hyphen).

Package names (including namespace packages) are also permitted. When a package name is supplied instead of a normal module, the interpreter will execute `<pkg>.__main__` as the main module. This behaviour is deliberately similar to the handling of directories and zipfiles that are passed to the interpreter as the script argument.

Note: This option cannot be used with built-in modules and extension modules written in C, since they do not have Python module files. However, it can still be used for precompiled modules, even if the original source file is not available.

If this option is given, the first element of `sys.argv` will be the full path to the module file (while the module file is being located, the first element will be set to `"-m"`). As with the *-c* option, the current directory will be added to the start of `sys.path`.

-I option can be used to run the script in isolated mode where `sys.path` contains neither the current directory nor the user's site-packages directory. All `PYTHON*` environment variables are ignored, too.

Many standard library modules contain code that is invoked on their execution as a script. An example is the `timeit` module:

```
python -m timeit -s "setup here" "benchmarked code here"
python -m timeit -h # for details
```

Raises an auditing event `cpython.run_module` with argument `module-name`.

See also:

runpy.run_module()
Equivalent functionality directly available to Python code

PEP 338 – Executing modules as scripts

Changed in version 3.1: Supply the package name to run a `__main__` submodule.

Changed in version 3.4: namespace packages are also supported

-

Read commands from standard input (`sys.stdin`). If standard input is a terminal, *-i* is implied.

If this option is given, the first element of `sys.argv` will be `"-"` and the current directory will be added to the start of `sys.path`.

Raises an auditing event `cpython.run_stdin` with no arguments.

<script>

Execute the Python code contained in *script*, which must be a filesystem path (absolute or relative) referring to either a Python file, a directory containing a `__main__.py` file, or a zipfile containing a `__main__.py` file.

If this option is given, the first element of `sys.argv` will be the script name as given on the command line.

If the script name refers directly to a Python file, the directory containing that file is added to the start of `sys.path`, and the file is executed as the `__main__` module.

If the script name refers to a directory or zipfile, the script name is added to the start of `sys.path` and the `__main__.py` file in that location is executed as the `__main__` module.

-I option can be used to run the script in isolated mode where `sys.path` contains neither the script's directory nor the user's site-packages directory. All `PYTHON*` environment variables are ignored, too.

Raises an auditing event `cpython.run_file` with argument `filename`.

See also:

runpy.run_path()
Equivalent functionality directly available to Python code

If no interface option is given, *-i* is implied, `sys.argv[0]` is an empty string (`""`) and the current directory will be added to the start of `sys.path`. Also, tab-completion and history editing is automatically enabled, if available on your platform (see rlcompleter-config).

See also:

tut-invoking

Changed in version 3.4: Automatic enabling of tab-completion and history editing.

1.1.2 Generic options

-?

-h

--help

Print a short description of all command line options and corresponding environment variables and exit.

--help-env

Print a short description of Python-specific environment variables and exit.

Added in version 3.11.

--help-xoptions

Print a description of implementation-specific *-X* options and exit.

Added in version 3.11.

--help-all

Print complete usage information and exit.

Added in version 3.11.

-V

--version

Print the Python version number and exit. Example output could be:

```
Python 3.8.0b2+
```

When given twice, print more information about the build, like:

```
Python 3.8.0b2+ (3.8:0c076caaa8, Apr 20 2019, 21:55:00)
[GCC 6.2.0 20161005]
```

Added in version 3.6: The `-VV` option.

1.1.3 Miscellaneous options

-b

Issue a warning when converting `bytes` or `bytearray` to `str` without specifying encoding or comparing `bytes` or `bytearray` with `str` or `bytes` with `int`. Issue an error when the option is given twice (`-bb`).

Changed in version 3.5: Affects also comparisons of `bytes` with `int`.

-B

If given, Python won't try to write `.pyc` files on the import of source modules. See also *PYTHONDONTWRITEBYTECODE*.

--check-hash-based-pycs `default|always|never`

Control the validation behavior of hash-based `.pyc` files. See pyc-invalidation. When set to `default`, checked and unchecked hash-based bytecode cache files are validated according to their default semantics. When set to `always`, all hash-based `.pyc` files, whether checked or unchecked, are validated against their corresponding source file. When set to `never`, hash-based `.pyc` files are not validated against their corresponding source files.

The semantics of timestamp-based `.pyc` files are unaffected by this option.

-d

Turn on parser debugging output (for expert only). See also the *PYTHONDEBUG* environment variable.

This option requires a *debug build of Python*, otherwise it's ignored.

-E

Ignore all `PYTHON*` environment variables, e.g. *PYTHONPATH* and *PYTHONHOME*, that might be set.

See also the *-P* and *-I* (isolated) options.

-i

When a script is passed as first argument or the *-c* option is used, enter interactive mode after executing the script or the command, even when `sys.stdin` does not appear to be a terminal. The *PYTHONSTARTUP* file is not read.

This can be useful to inspect global variables or a stack trace when a script raises an exception. See also *PYTHONINSPECT*.

-I

Run Python in isolated mode. This also implies *-E*, *-P* and *-s* options.

In isolated mode `sys.path` contains neither the script's directory nor the user's site-packages directory. All `PYTHON*` environment variables are ignored, too. Further restrictions may be imposed to prevent the user from injecting malicious code.

Added in version 3.4.

-O

Remove assert statements and any code conditional on the value of `__debug__`. Augment the filename for compiled (*bytecode*) files by adding `.opt-1` before the `.pyc` extension (see **PEP 488**). See also `PYTHONOPTIMIZE`.

Changed in version 3.5: Modify `.pyc` filenames according to **PEP 488**.

-OO

Do `-O` and also discard docstrings. Augment the filename for compiled (*bytecode*) files by adding `.opt-2` before the `.pyc` extension (see **PEP 488**).

Changed in version 3.5: Modify `.pyc` filenames according to **PEP 488**.

-P

Don't prepend a potentially unsafe path to `sys.path`:

- `python -m module` command line: Don't prepend the current working directory.
- `python script.py` command line: Don't prepend the script's directory. If it's a symbolic link, resolve symbolic links.
- `python -c code` and `python` (REPL) command lines: Don't prepend an empty string, which means the current working directory.

See also the `PYTHONSAFEPATH` environment variable, and `-E` and `-I` (isolated) options.

Added in version 3.11.

-q

Don't display the copyright and version messages even in interactive mode.

Added in version 3.2.

-R

Turn on hash randomization. This option only has an effect if the `PYTHONHASHSEED` environment variable is set to `0`, since hash randomization is enabled by default.

On previous versions of Python, this option turns on hash randomization, so that the `__hash__()` values of str and bytes objects are "salted" with an unpredictable random value. Although they remain constant within an individual Python process, they are not predictable between repeated invocations of Python.

Hash randomization is intended to provide protection against a denial-of-service caused by carefully chosen inputs that exploit the worst case performance of a dict construction, $O(n^2)$ complexity. See http://ocert.org/advisories/ocert-2011-003.html for details.

`PYTHONHASHSEED` allows you to set a fixed value for the hash seed secret.

Added in version 3.2.3.

Changed in version 3.7: The option is no longer ignored.

-s

Don't add the `user site-packages directory` to `sys.path`.

See also `PYTHONNOUSERSITE`.

See also:

PEP 370 – Per user site-packages directory

-S

Disable the import of the module `site` and the site-dependent manipulations of `sys.path` that it entails. Also disable these manipulations if `site` is explicitly imported later (call `site.main()` if you want them to be triggered).

-u

Force the stdout and stderr streams to be unbuffered. This option has no effect on the stdin stream.

See also *PYTHONUNBUFFERED*.

Changed in version 3.7: The text layer of the stdout and stderr streams now is unbuffered.

-v

Print a message each time a module is initialized, showing the place (filename or built-in module) from which it is loaded. When given twice (`-vv`), print a message for each file that is checked for when searching for a module. Also provides information on module cleanup at exit.

Changed in version 3.10: The `site` module reports the site-specific paths and `.pth` files being processed.

See also *PYTHONVERBOSE*.

-W `arg`

Warning control. Python's warning machinery by default prints warning messages to `sys.stderr`.

The simplest settings apply a particular action unconditionally to all warnings emitted by a process (even those that are otherwise ignored by default):

```
-Wdefault  # Warn once per call location
-Werror    # Convert to exceptions
-Walways   # Warn every time
-Wall      # Same as -Walways
-Wmodule   # Warn once per calling module
-Wonce     # Warn once per Python process
-Wignore   # Never warn
```

The action names can be abbreviated as desired and the interpreter will resolve them to the appropriate action name. For example, `-Wi` is the same as `-Wignore`.

The full form of argument is:

```
action:message:category:module:lineno
```

Empty fields match all values; trailing empty fields may be omitted. For example `-W ignore::DeprecationWarning` ignores all DeprecationWarning warnings.

The *action* field is as explained above but only applies to warnings that match the remaining fields.

The *message* field must match the whole warning message; this match is case-insensitive.

The *category* field matches the warning category (ex: `DeprecationWarning`). This must be a class name; the match test whether the actual warning category of the message is a subclass of the specified warning category.

The *module* field matches the (fully qualified) module name; this match is case-sensitive.

The *lineno* field matches the line number, where zero matches all line numbers and is thus equivalent to an omitted line number.

Multiple *-W* options can be given; when a warning matches more than one option, the action for the last matching option is performed. Invalid *-W* options are ignored (though, a warning message is printed about invalid options when the first warning is issued).

Warnings can also be controlled using the *PYTHONWARNINGS* environment variable and from within a Python program using the `warnings` module. For example, the `warnings.filterwarnings()` function can be used to use a regular expression on the warning message.

See warning-filter and describing-warning-filters for more details.

-x

Skip the first line of the source, allowing use of non-Unix forms of `#!cmd`. This is intended for a DOS specific hack only.

-X

Reserved for various implementation-specific options. CPython currently defines the following possible values:

- `-X faulthandler` to enable `faulthandler`. See also *PYTHONFAULTHANDLER*.

  Added in version 3.3.

- `-X showrefcount` to output the total reference count and number of used memory blocks when the program finishes or after each statement in the interactive interpreter. This only works on *debug builds*.

  Added in version 3.4.

- `-X tracemalloc` to start tracing Python memory allocations using the `tracemalloc` module. By default, only the most recent frame is stored in a traceback of a trace. Use `-X tracemalloc=NFRAME` to start tracing with a traceback limit of *NFRAME* frames. See `tracemalloc.start()` and *PYTHONTRACEMALLOC* for more information.

  Added in version 3.4.

- `-X int_max_str_digits` configures the integer string conversion length limitation. See also *PYTHONINTMAXSTRDIGITS*.

  Added in version 3.11.

- `-X importtime` to show how long each import takes. It shows module name, cumulative time (including nested imports) and self time (excluding nested imports). Note that its output may be broken in multi-threaded application. Typical usage is `python3 -X importtime -c 'import asyncio'`. See also *PYTHONPROFILEIMPORTTIME*.

  Added in version 3.7.

- `-X dev`: enable Python Development Mode, introducing additional runtime checks that are too expensive to be enabled by default. See also *PYTHONDEVMODE*.

  Added in version 3.7.

- `-X utf8` enables the Python UTF-8 Mode. `-X utf8=0` explicitly disables Python UTF-8 Mode (even when it would otherwise activate automatically). See also *PYTHONUTF8*.

  Added in version 3.7.

- `-X pycache_prefix=PATH` enables writing `.pyc` files to a parallel tree rooted at the given directory instead of to the code tree. See also *PYTHONPYCACHEPREFIX*.

  Added in version 3.8.

- `-X warn_default_encoding` issues a `EncodingWarning` when the locale-specific default encoding is used for opening files. See also *PYTHONWARNDEFAULTENCODING*.

  Added in version 3.10.

- `-X no_debug_ranges` disables the inclusion of the tables mapping extra location information (end line, start column offset and end column offset) to every instruction in code objects. This is useful when smaller code objects and pyc files are desired as well as suppressing the extra visual location indicators when the interpreter displays tracebacks. See also *PYTHONNODEBUGRANGES*.

  Added in version 3.11.

- `-X frozen_modules` determines whether or not frozen modules are ignored by the import machinery. A value of "on" means they get imported and "off" means they are ignored. The default is "on" if this is an

installed Python (the normal case). If it's under development (running from the source tree) then the default is "off". Note that the "importlib_bootstrap" and "importlib_bootstrap_external" frozen modules are always used, even if this flag is set to "off".

Added in version 3.11.

- `-X perf` enables support for the Linux `perf` profiler. When this option is provided, the `perf` profiler will be able to report Python calls. This option is only available on some platforms and will do nothing if is not supported on the current system. The default value is "off". See also *PYTHONPERFSUPPORT* and perf_profiling.

  Added in version 3.12.

It also allows passing arbitrary values and retrieving them through the `sys._xoptions` dictionary.

Added in version 3.2.

Changed in version 3.9: Removed the `-X showalloccount` option.

Changed in version 3.10: Removed the `-X oldparser` option.

1.1.4 Options you shouldn't use

-J

Reserved for use by Jython.

1.2 Environment variables

These environment variables influence Python's behavior, they are processed before the command-line switches other than -E or -I. It is customary that command-line switches override environmental variables where there is a conflict.

PYTHONHOME

Change the location of the standard Python libraries. By default, the libraries are searched in *prefix*`/lib/``python`*version* and *exec_prefix*`/lib/python`*version*, where *prefix* and *exec_prefix* are installation-dependent directories, both defaulting to `/usr/local`.

When *PYTHONHOME* is set to a single directory, its value replaces both *prefix* and *exec_prefix*. To specify different values for these, set *PYTHONHOME* to *prefix*`:`*exec_prefix*.

PYTHONPATH

Augment the default search path for module files. The format is the same as the shell's `PATH`: one or more directory pathnames separated by `os.pathsep` (e.g. colons on Unix or semicolons on Windows). Non-existent directories are silently ignored.

In addition to normal directories, individual *PYTHONPATH* entries may refer to zipfiles containing pure Python modules (in either source or compiled form). Extension modules cannot be imported from zipfiles.

The default search path is installation dependent, but generally begins with *prefix*`/lib/python`*version* (see *PYTHONHOME* above). It is *always* appended to *PYTHONPATH*.

An additional directory will be inserted in the search path in front of *PYTHONPATH* as described above under *Interface options*. The search path can be manipulated from within a Python program as the variable `sys.path`.

PYTHONSAFEPATH

If this is set to a non-empty string, don't prepend a potentially unsafe path to `sys.path`: see the *-P* option for details.

Added in version 3.11.

PYTHONPLATLIBDIR

If this is set to a non-empty string, it overrides the `sys.platlibdir` value.

Added in version 3.9.

PYTHONSTARTUP

If this is the name of a readable file, the Python commands in that file are executed before the first prompt is displayed in interactive mode. The file is executed in the same namespace where interactive commands are executed so that objects defined or imported in it can be used without qualification in the interactive session. You can also change the prompts `sys.ps1` and `sys.ps2` and the hook `sys.__interactivehook__` in this file.

Raises an auditing event `cpython.run_startup` with the filename as the argument when called on startup.

PYTHONOPTIMIZE

If this is set to a non-empty string it is equivalent to specifying the *-O* option. If set to an integer, it is equivalent to specifying *-O* multiple times.

PYTHONBREAKPOINT

If this is set, it names a callable using dotted-path notation. The module containing the callable will be imported and then the callable will be run by the default implementation of `sys.breakpointhook()` which itself is called by built-in `breakpoint()`. If not set, or set to the empty string, it is equivalent to the value "pdb.set_trace". Setting this to the string "0" causes the default implementation of `sys.breakpointhook()` to do nothing but return immediately.

Added in version 3.7.

PYTHONDEBUG

If this is set to a non-empty string it is equivalent to specifying the *-d* option. If set to an integer, it is equivalent to specifying *-d* multiple times.

This environment variable requires a *debug build of Python*, otherwise it's ignored.

PYTHONINSPECT

If this is set to a non-empty string it is equivalent to specifying the *-i* option.

This variable can also be modified by Python code using `os.environ` to force inspect mode on program termination.

PYTHONUNBUFFERED

If this is set to a non-empty string it is equivalent to specifying the *-u* option.

PYTHONVERBOSE

If this is set to a non-empty string it is equivalent to specifying the *-v* option. If set to an integer, it is equivalent to specifying *-v* multiple times.

PYTHONCASEOK

If this is set, Python ignores case in `import` statements. This only works on Windows and macOS.

PYTHONDONTWRITEBYTECODE

If this is set to a non-empty string, Python won't try to write `.pyc` files on the import of source modules. This is equivalent to specifying the *-B* option.

PYTHONPYCACHEPREFIX

If this is set, Python will write `.pyc` files in a mirror directory tree at this path, instead of in `__pycache__` directories within the source tree. This is equivalent to specifying the *-X* `pycache_prefix=PATH` option.

Added in version 3.8.

PYTHONHASHSEED

If this variable is not set or set to `random`, a random value is used to seed the hashes of str and bytes objects.

If *PYTHONHASHSEED* is set to an integer value, it is used as a fixed seed for generating the hash() of the types covered by the hash randomization.

Its purpose is to allow repeatable hashing, such as for selftests for the interpreter itself, or to allow a cluster of python processes to share hash values.

The integer must be a decimal number in the range [0,4294967295]. Specifying the value 0 will disable hash randomization.

Added in version 3.2.3.

PYTHONINTMAXSTRDIGITS

If this variable is set to an integer, it is used to configure the interpreter's global integer string conversion length limitation.

Added in version 3.11.

PYTHONIOENCODING

If this is set before running the interpreter, it overrides the encoding used for stdin/stdout/stderr, in the syntax `encodingname:errorhandler`. Both the `encodingname` and the `:errorhandler` parts are optional and have the same meaning as in `str.encode()`.

For stderr, the `:errorhandler` part is ignored; the handler will always be `'backslashreplace'`.

Changed in version 3.4: The `encodingname` part is now optional.

Changed in version 3.6: On Windows, the encoding specified by this variable is ignored for interactive console buffers unless *PYTHONLEGACYWINDOWSSTDIO* is also specified. Files and pipes redirected through the standard streams are not affected.

PYTHONNOUSERSITE

If this is set, Python won't add the `user site-packages directory` to `sys.path`.

See also:

PEP 370 – Per user site-packages directory

PYTHONUSERBASE

Defines the `user base directory`, which is used to compute the path of the `user site-packages directory` and installation paths for `python -m pip install --user`.

See also:

PEP 370 – Per user site-packages directory

PYTHONEXECUTABLE

If this environment variable is set, `sys.argv[0]` will be set to its value instead of the value got through the C runtime. Only works on macOS.

PYTHONWARNINGS

This is equivalent to the *-W* option. If set to a comma separated string, it is equivalent to specifying *-W* multiple times, with filters later in the list taking precedence over those earlier in the list.

The simplest settings apply a particular action unconditionally to all warnings emitted by a process (even those that are otherwise ignored by default):

```
PYTHONWARNINGS=default  # Warn once per call location
PYTHONWARNINGS=error    # Convert to exceptions
PYTHONWARNINGS=always   # Warn every time
PYTHONWARNINGS=all      # Same as PYTHONWARNINGS=always
PYTHONWARNINGS=module   # Warn once per calling module
PYTHONWARNINGS=once     # Warn once per Python process
PYTHONWARNINGS=ignore   # Never warn
```

See warning-filter and describing-warning-filters for more details.

PYTHONFAULTHANDLER

If this environment variable is set to a non-empty string, `faulthandler.enable()` is called at startup: install a handler for `SIGSEGV`, `SIGFPE`, `SIGABRT`, `SIGBUS` and `SIGILL` signals to dump the Python traceback. This is equivalent to *-X* `faulthandler` option.

Added in version 3.3.

PYTHONTRACEMALLOC

If this environment variable is set to a non-empty string, start tracing Python memory allocations using the `tracemalloc` module. The value of the variable is the maximum number of frames stored in a traceback of a trace. For example, `PYTHONTRACEMALLOC=1` stores only the most recent frame. See the `tracemalloc.start()` function for more information. This is equivalent to setting the *-X* `tracemalloc` option.

Added in version 3.4.

PYTHONPROFILEIMPORTTIME

If this environment variable is set to a non-empty string, Python will show how long each import takes. This is equivalent to setting the *-X* `importtime` option.

Added in version 3.7.

PYTHONASYNCIODEBUG

If this environment variable is set to a non-empty string, enable the debug mode of the `asyncio` module.

Added in version 3.4.

PYTHONMALLOC

Set the Python memory allocators and/or install debug hooks.

Set the family of memory allocators used by Python:

- `default`: use the default memory allocators.
- `malloc`: use the `malloc()` function of the C library for all domains (`PYMEM_DOMAIN_RAW`, `PYMEM_DOMAIN_MEM`, `PYMEM_DOMAIN_OBJ`).
- `pymalloc`: use the pymalloc allocator for `PYMEM_DOMAIN_MEM` and `PYMEM_DOMAIN_OBJ` domains and use the `malloc()` function for the `PYMEM_DOMAIN_RAW` domain.

Install debug hooks:

- `debug`: install debug hooks on top of the default memory allocators.
- `malloc_debug`: same as `malloc` but also install debug hooks.
- `pymalloc_debug`: same as `pymalloc` but also install debug hooks.

Added in version 3.6.

Changed in version 3.7: Added the `"default"` allocator.

PYTHONMALLOCSTATS

If set to a non-empty string, Python will print statistics of the pymalloc memory allocator every time a new pymalloc object arena is created, and on shutdown.

This variable is ignored if the *PYTHONMALLOC* environment variable is used to force the `malloc()` allocator of the C library, or if Python is configured without `pymalloc` support.

Changed in version 3.6: This variable can now also be used on Python compiled in release mode. It now has no effect if set to an empty string.

PYTHONLEGACYWINDOWSFSENCODING

If set to a non-empty string, the default *filesystem encoding and error handler* mode will revert to their pre-3.6 values of 'mbcs' and 'replace', respectively. Otherwise, the new defaults 'utf-8' and 'surrogatepass' are used.

This may also be enabled at runtime with `sys._enablelegacywindowsfsencoding()`.

Availability: Windows.

Added in version 3.6: See **PEP 529** for more details.

PYTHONLEGACYWINDOWSSTDIO

If set to a non-empty string, does not use the new console reader and writer. This means that Unicode characters will be encoded according to the active console code page, rather than using utf-8.

This variable is ignored if the standard streams are redirected (to files or pipes) rather than referring to console buffers.

Availability: Windows.

Added in version 3.6.

PYTHONCOERCECLOCALE

If set to the value `0`, causes the main Python command line application to skip coercing the legacy ASCII-based C and POSIX locales to a more capable UTF-8 based alternative.

If this variable is *not* set (or is set to a value other than `0`), the `LC_ALL` locale override environment variable is also not set, and the current locale reported for the `LC_CTYPE` category is either the default `C` locale, or else the explicitly ASCII-based `POSIX` locale, then the Python CLI will attempt to configure the following locales for the `LC_CTYPE` category in the order listed before loading the interpreter runtime:

- `C.UTF-8`
- `C.utf8`
- `UTF-8`

If setting one of these locale categories succeeds, then the `LC_CTYPE` environment variable will also be set accordingly in the current process environment before the Python runtime is initialized. This ensures that in addition to being seen by both the interpreter itself and other locale-aware components running in the same process (such as the GNU `readline` library), the updated setting is also seen in subprocesses (regardless of whether or not those processes are running a Python interpreter), as well as in operations that query the environment rather than the current C locale (such as Python's own `locale.getdefaultlocale()`).

Configuring one of these locales (either explicitly or via the above implicit locale coercion) automatically enables the `surrogateescape` error handler for `sys.stdin` and `sys.stdout` (`sys.stderr` continues to use `backslashreplace` as it does in any other locale). This stream handling behavior can be overridden using *PYTHONIOENCODING* as usual.

For debugging purposes, setting `PYTHONCOERCECLOCALE=warn` will cause Python to emit warning messages on `stderr` if either the locale coercion activates, or else if a locale that *would* have triggered coercion is still active when the Python runtime is initialized.

Also note that even when locale coercion is disabled, or when it fails to find a suitable target locale, *PYTHONUTF8* will still activate by default in legacy ASCII-based locales. Both features must be disabled in order to force the interpreter to use `ASCII` instead of `UTF-8` for system interfaces.

Availability: Unix.

Added in version 3.7: See **PEP 538** for more details.

PYTHONDEVMODE

If this environment variable is set to a non-empty string, enable Python Development Mode, introducing additional runtime checks that are too expensive to be enabled by default. This is equivalent to setting the *-X* `dev` option.

Added in version 3.7.

PYTHONUTF8

If set to `1`, enable the Python UTF-8 Mode.

If set to `0`, disable the Python UTF-8 Mode.

Setting any other non-empty string causes an error during interpreter initialisation.

Added in version 3.7.

PYTHONWARNDEFAULTENCODING

If this environment variable is set to a non-empty string, issue a `EncodingWarning` when the locale-specific default encoding is used.

See io-encoding-warning for details.

Added in version 3.10.

PYTHONNODEBUGRANGES

If this variable is set, it disables the inclusion of the tables mapping extra location information (end line, start column offset and end column offset) to every instruction in code objects. This is useful when smaller code objects and pyc files are desired as well as suppressing the extra visual location indicators when the interpreter displays tracebacks.

Added in version 3.11.

PYTHONPERFSUPPORT

If this variable is set to a nonzero value, it enables support for the Linux `perf` profiler so Python calls can be detected by it.

If set to `0`, disable Linux `perf` profiler support.

See also the *-X perf* command-line option and perf_profiling.

Added in version 3.12.

1.2.1 Debug-mode variables

PYTHONDUMPREFS

If set, Python will dump objects and reference counts still alive after shutting down the interpreter.

Need Python configured with the *--with-trace-refs* build option.

PYTHONDUMPREFSFILE=FILENAME

If set, Python will dump objects and reference counts still alive after shutting down the interpreter into a file called *FILENAME*.

Need Python configured with the *--with-trace-refs* build option.

Added in version 3.11.

CHAPTER

TWO

USING PYTHON ON UNIX PLATFORMS

2.1 Getting and installing the latest version of Python

2.1.1 On Linux

Python comes preinstalled on most Linux distributions, and is available as a package on all others. However there are certain features you might want to use that are not available on your distro's package. You can easily compile the latest version of Python from source.

In the event that Python doesn't come preinstalled and isn't in the repositories as well, you can easily make packages for your own distro. Have a look at the following links:

See also:

https://www.debian.org/doc/manuals/maint-guide/first.en.html
for Debian users

https://en.opensuse.org/Portal:Packaging
for OpenSuse users

https://docs.fedoraproject.org/en-US/package-maintainers/Packaging_Tutorial_GNU_Hello/
for Fedora users

https://slackbook.org/html/package-management-making-packages.html
for Slackware users

2.1.2 On FreeBSD and OpenBSD

- FreeBSD users, to add the package use:

```
pkg install python3
```

- OpenBSD users, to add the package use:

```
pkg_add -r python

pkg_add ftp://ftp.openbsd.org/pub/OpenBSD/4.2/packages/<insert your architecture␣
↪here>/python-<version>.tgz
```

For example i386 users get the 2.5.1 version of Python using:

```
pkg_add ftp://ftp.openbsd.org/pub/OpenBSD/4.2/packages/i386/python-2.5.1p2.tgz
```

2.2 Building Python

If you want to compile CPython yourself, first thing you should do is get the source. You can download either the latest release's source or just grab a fresh clone. (If you want to contribute patches, you will need a clone.)

The build process consists of the usual commands:

```
./configure
make
make install
```

Configuration options and caveats for specific Unix platforms are extensively documented in the README.rst file in the root of the Python source tree.

> **Warning:** `make install` can overwrite or masquerade the `python3` binary. `make altinstall` is therefore recommended instead of `make install` since it only installs *exec_prefix*/bin/python*version*.

2.3 Python-related paths and files

These are subject to difference depending on local installation conventions; *prefix* and *exec_prefix* are installation-dependent and should be interpreted as for GNU software; they may be the same.

For example, on most Linux systems, the default for both is `/usr`.

File/directory	Meaning
exec_prefix/bin/python3	Recommended location of the interpreter.
prefix/lib/python*version*, *exec_prefix*/lib/python*version*	Recommended locations of the directories containing the standard modules.
prefix/include/python*version*, *exec_prefix*/include/python*version*	Recommended locations of the directories containing the include files needed for developing Python extensions and embedding the interpreter.

2.4 Miscellaneous

To easily use Python scripts on Unix, you need to make them executable, e.g. with

```
$ chmod +x script
```

and put an appropriate Shebang line at the top of the script. A good choice is usually

```
#!/usr/bin/env python3
```

which searches for the Python interpreter in the whole `PATH`. However, some Unices may not have the **env** command, so you may need to hardcode `/usr/bin/python3` as the interpreter path.

To use shell commands in your Python scripts, look at the `subprocess` module.

2.5 Custom OpenSSL

1. To use your vendor's OpenSSL configuration and system trust store, locate the directory with `openssl.cnf` file or symlink in `/etc`. On most distribution the file is either in `/etc/ssl` or `/etc/pki/tls`. The directory should also contain a `cert.pem` file and/or a `certs` directory.

```
$ find /etc/ -name openssl.cnf -printf "%h\n"
/etc/ssl
```

2. Download, build, and install OpenSSL. Make sure you use `install_sw` and not `install`. The `install_sw` target does not override `openssl.cnf`.

```
$ curl -O https://www.openssl.org/source/openssl-VERSION.tar.gz
$ tar xzf openssl-VERSION
$ pushd openssl-VERSION
$ ./config \
    --prefix=/usr/local/custom-openssl \
    --libdir=lib \
    --openssldir=/etc/ssl
$ make -j1 depend
$ make -j8
$ make install_sw
$ popd
```

3. Build Python with custom OpenSSL (see the configure `--with-openssl` and `--with-openssl-rpath` options)

```
$ pushd python-3.x.x
$ ./configure -C \
    --with-openssl=/usr/local/custom-openssl \
    --with-openssl-rpath=auto \
    --prefix=/usr/local/python-3.x.x
$ make -j8
$ make altinstall
```

Note: Patch releases of OpenSSL have a backwards compatible ABI. You don't need to recompile Python to update OpenSSL. It's sufficient to replace the custom OpenSSL installation with a newer version.

CHAPTER

THREE

CONFIGURE PYTHON

3.1 Build Requirements

Features required to build CPython:

- A C11 compiler. Optional C11 features are not required.
- Support for IEEE 754 floating point numbers and floating point Not-a-Number (NaN).
- Support for threads.
- OpenSSL 1.1.1 or newer for the `ssl` and `hashlib` modules.
- On Windows, Microsoft Visual Studio 2017 or later is required.

Changed in version 3.5: On Windows, Visual Studio 2015 or later is required.

Changed in version 3.6: Selected C99 features are now required, like `<stdint.h>` and `static inline` functions.

Changed in version 3.7: Thread support and OpenSSL 1.0.2 are now required.

Changed in version 3.10: OpenSSL 1.1.1 is now required.

Changed in version 3.11: C11 compiler, IEEE 754 and NaN support are now required. On Windows, Visual Studio 2017 or later is required.

See also **PEP 7** "Style Guide for C Code" and **PEP 11** "CPython platform support".

3.2 Generated files

To reduce build dependencies, Python source code contains multiple generated files. Commands to regenerate all generated files:

```
make regen-all
make regen-stdlib-module-names
make regen-limited-abi
make regen-configure
```

The `Makefile.pre.in` file documents generated files, their inputs, and tools used to regenerate them. Search for `regen-*` make targets.

3.2.1 configure script

The `make regen-configure` command regenerates the `aclocal.m4` file and the `configure` script using the `Tools/build/regen-configure.sh` shell script which uses an Ubuntu container to get the same tools versions and have a reproducible output.

The container is optional, the following command can be run locally:

```
autoreconf -ivf -Werror
```

The generated files can change depending on the exact `autoconf-archive`, `aclocal` and `pkg-config` versions.

3.3 Configure Options

List all `./configure` script options using:

```
./configure --help
```

See also the `Misc/SpecialBuilds.txt` in the Python source distribution.

3.3.1 General Options

--enable-loadable-sqlite-extensions

Support loadable extensions in the `_sqlite` extension module (default is no) of the `sqlite3` module.

See the `sqlite3.Connection.enable_load_extension()` method of the `sqlite3` module.

Added in version 3.6.

--disable-ipv6

Disable IPv6 support (enabled by default if supported), see the `socket` module.

--enable-big-digits=`[15|30]`

Define the size in bits of Python `int` digits: 15 or 30 bits.

By default, the digit size is 30.

Define the `PYLONG_BITS_IN_DIGIT` to `15` or `30`.

See `sys.int_info.bits_per_digit`.

--with-suffix=`SUFFIX`

Set the Python executable suffix to *SUFFIX*.

The default suffix is `.exe` on Windows and macOS (`python.exe` executable), `.js` on Emscripten node, `.html` on Emscripten browser, `.wasm` on WASI, and an empty string on other platforms (`python` executable).

Changed in version 3.11: The default suffix on WASM platform is one of `.js`, `.html` or `.wasm`.

--with-tzpath=`<list of absolute paths separated by pathsep>`

Select the default time zone search path for `zoneinfo.TZPATH`. See the Compile-time configuration of the `zoneinfo` module.

Default: `/usr/share/zoneinfo:/usr/lib/zoneinfo:/usr/share/lib/zoneinfo:/etc/zoneinfo`.

See `os.pathsep` path separator.

Added in version 3.9.

--without-decimal-contextvar

Build the `_decimal` extension module using a thread-local context rather than a coroutine-local context (default), see the `decimal` module.

See `decimal.HAVE_CONTEXTVAR` and the `contextvars` module.

Added in version 3.9.

--with-dbmliborder=`<list of backend names>`

Override order to check db backends for the `dbm` module

A valid value is a colon (`:`) separated string with the backend names:

- `ndbm`;
- `gdbm`;
- `bdb`.

--without-c-locale-coercion

Disable C locale coercion to a UTF-8 based locale (enabled by default).

Don't define the `PY_COERCE_C_LOCALE` macro.

See *`PYTHONCOERCECLOCALE`* and the **PEP 538**.

--without-freelists

Disable all freelists except the empty tuple singleton.

Added in version 3.11.

--with-platlibdir=`DIRNAME`

Python library directory name (default is `lib`).

Fedora and SuSE use `lib64` on 64-bit platforms.

See `sys.platlibdir`.

Added in version 3.9.

--with-wheel-pkg-dir=`PATH`

Directory of wheel packages used by the `ensurepip` module (none by default).

Some Linux distribution packaging policies recommend against bundling dependencies. For example, Fedora installs wheel packages in the `/usr/share/python-wheels/` directory and don't install the `ensurepip._bundled` package.

Added in version 3.10.

--with-pkg-config=`[check|yes|no]`

Whether configure should use **pkg-config** to detect build dependencies.

- `check` (default): **pkg-config** is optional
- `yes`: **pkg-config** is mandatory
- `no`: configure does not use **pkg-config** even when present

Added in version 3.11.

--enable-pystats

Turn on internal statistics gathering.

The statistics will be dumped to a arbitrary (probably unique) file in `/tmp/py_stats/`, or `C:\temp\py_stats\` on Windows. If that directory does not exist, results will be printed on stdout.

Use `Tools/scripts/summarize_stats.py` to read the stats.

Added in version 3.11.

3.3.2 WebAssembly Options

--with-emscripten-target=`[browser|node]`

Set build flavor for `wasm32-emscripten`.

- `browser` (default): preload minimal stdlib, default MEMFS.
- `node`: NODERAWFS and pthread support.

Added in version 3.11.

--enable-wasm-dynamic-linking

Turn on dynamic linking support for WASM.

Dynamic linking enables `dlopen`. File size of the executable increases due to limited dead code elimination and additional features.

Added in version 3.11.

--enable-wasm-pthreads

Turn on pthreads support for WASM.

Added in version 3.11.

3.3.3 Install Options

--prefix=`PREFIX`

Install architecture-independent files in PREFIX. On Unix, it defaults to `/usr/local`.

This value can be retrieved at runtime using `sys.prefix`.

As an example, one can use `--prefix="$HOME/.local/"` to install a Python in its home directory.

--exec-prefix=`EPREFIX`

Install architecture-dependent files in EPREFIX, defaults to *--prefix*.

This value can be retrieved at runtime using `sys.exec_prefix`.

--disable-test-modules

Don't build nor install test modules, like the `test` package or the `_testcapi` extension module (built and installed by default).

Added in version 3.10.

--with-ensurepip=`[upgrade|install|no]`

Select the `ensurepip` command run on Python installation:

- `upgrade` (default): run `python -m ensurepip --altinstall --upgrade` command.
- `install`: run `python -m ensurepip --altinstall` command;

- `no`: don't run ensurepip;

Added in version 3.6.

3.3.4 Performance options

Configuring Python using `--enable-optimizations --with-lto` (PGO + LTO) is recommended for best performance. The experimental `--enable-bolt` flag can also be used to improve performance.

`--enable-optimizations`

Enable Profile Guided Optimization (PGO) using *`PROFILE_TASK`* (disabled by default).

The C compiler Clang requires `llvm-profdata` program for PGO. On macOS, GCC also requires it: GCC is just an alias to Clang on macOS.

Disable also semantic interposition in libpython if `--enable-shared` and GCC is used: add `-fno-semantic-interposition` to the compiler and linker flags.

Note: During the build, you may encounter compiler warnings about profile data not being available for some source files. These warnings are harmless, as only a subset of the code is exercised during profile data acquisition. To disable these warnings on Clang, manually suppress them by adding `-Wno-profile-instr-unprofiled` to *`CFLAGS`*.

Added in version 3.6.

Changed in version 3.10: Use `-fno-semantic-interposition` on GCC.

`PROFILE_TASK`

Environment variable used in the Makefile: Python command line arguments for the PGO generation task.

Default: `-m test --pgo --timeout=$(TESTTIMEOUT)`.

Added in version 3.8.

`--with-lto``=[full|thin|no|yes]`

Enable Link Time Optimization (LTO) in any build (disabled by default).

The C compiler Clang requires `llvm-ar` for LTO (`ar` on macOS), as well as an LTO-aware linker (`ld.gold` or `lld`).

Added in version 3.6.

Added in version 3.11: To use ThinLTO feature, use `--with-lto=thin` on Clang.

Changed in version 3.12: Use ThinLTO as the default optimization policy on Clang if the compiler accepts the flag.

`--enable-bolt`

Enable usage of the BOLT post-link binary optimizer (disabled by default).

BOLT is part of the LLVM project but is not always included in their binary distributions. This flag requires that `llvm-bolt` and `merge-fdata` are available.

BOLT is still a fairly new project so this flag should be considered experimental for now. Because this tool operates on machine code its success is dependent on a combination of the build environment + the other optimization configure args + the CPU architecture, and not all combinations are supported. BOLT versions before LLVM 16 are known to crash BOLT under some scenarios. Use of LLVM 16 or newer for BOLT optimization is strongly encouraged.

The `BOLT_INSTRUMENT_FLAGS` and `BOLT_APPLY_FLAGS` **configure** variables can be defined to override the default set of arguments for **llvm-bolt** to instrument and apply BOLT data to binaries, respectively.

Added in version 3.12.

--with-computed-gotos

Enable computed gotos in evaluation loop (enabled by default on supported compilers).

--without-pymalloc

Disable the specialized Python memory allocator pymalloc (enabled by default).

See also *PYTHONMALLOC* environment variable.

--without-doc-strings

Disable static documentation strings to reduce the memory footprint (enabled by default). Documentation strings defined in Python are not affected.

Don't define the `WITH_DOC_STRINGS` macro.

See the `PyDoc_STRVAR()` macro.

--enable-profiling

Enable C-level code profiling with `gprof` (disabled by default).

--with-strict-overflow

Add `-fstrict-overflow` to the C compiler flags (by default we add `-fno-strict-overflow` instead).

3.3.5 Python Debug Build

A debug build is Python built with the *--with-pydebug* configure option.

Effects of a debug build:

- Display all warnings by default: the list of default warning filters is empty in the `warnings` module.
- Add `d` to `sys.abiflags`.
- Add `sys.gettotalrefcount()` function.
- Add *-X showrefcount* command line option.
- Add *-d* command line option and *PYTHONDEBUG* environment variable to debug the parser.
- Add support for the `__lltrace__` variable: enable low-level tracing in the bytecode evaluation loop if the variable is defined.
- Install debug hooks on memory allocators to detect buffer overflow and other memory errors.
- Define `Py_DEBUG` and `Py_REF_DEBUG` macros.
- Add runtime checks: code surrounded by `#ifdef Py_DEBUG` and `#endif`. Enable `assert(...)` and `_PyObject_ASSERT(...)` assertions: don't set the `NDEBUG` macro (see also the *--with-assertions* configure option). Main runtime checks:
  - Add sanity checks on the function arguments.
  - Unicode and int objects are created with their memory filled with a pattern to detect usage of uninitialized objects.
  - Ensure that functions which can clear or replace the current exception are not called with an exception raised.
  - Check that deallocator functions don't change the current exception.
  - The garbage collector (`gc.collect()` function) runs some basic checks on objects consistency.

- The `Py_SAFE_DOWNCAST()` macro checks for integer underflow and overflow when downcasting from wide types to narrow types.

See also the Python Development Mode and the `--with-trace-refs` configure option.

Changed in version 3.8: Release builds and debug builds are now ABI compatible: defining the `Py_DEBUG` macro no longer implies the `Py_TRACE_REFS` macro (see the `--with-trace-refs` option), which introduces the only ABI incompatibility.

3.3.6 Debug options

`--with-pydebug`

Build Python in debug mode: define the `Py_DEBUG` macro (disabled by default).

`--with-trace-refs`

Enable tracing references for debugging purpose (disabled by default).

Effects:

- Define the `Py_TRACE_REFS` macro.
- Add `sys.getobjects()` function.
- Add `PYTHONDUMPREFS` environment variable.

This build is not ABI compatible with release build (default build) or debug build (`Py_DEBUG` and `Py_REF_DEBUG` macros).

Added in version 3.8.

`--with-assertions`

Build with C assertions enabled (default is no): `assert(...);` and `_PyObject_ASSERT(...);`.

If set, the `NDEBUG` macro is not defined in the `OPT` compiler variable.

See also the `--with-pydebug` option (*debug build*) which also enables assertions.

Added in version 3.6.

`--with-valgrind`

Enable Valgrind support (default is no).

`--with-dtrace`

Enable DTrace support (default is no).

See Instrumenting CPython with DTrace and SystemTap.

Added in version 3.6.

`--with-address-sanitizer`

Enable AddressSanitizer memory error detector, `asan` (default is no).

Added in version 3.6.

`--with-memory-sanitizer`

Enable MemorySanitizer allocation error detector, `msan` (default is no).

Added in version 3.6.

`--with-undefined-behavior-sanitizer`

Enable UndefinedBehaviorSanitizer undefined behaviour detector, `ubsan` (default is no).

Added in version 3.6.

3.3.7 Linker options

--enable-shared

Enable building a shared Python library: `libpython` (default is no).

--without-static-libpython

Do not build `libpythonMAJOR.MINOR.a` and do not install `python.o` (built and enabled by default).

Added in version 3.10.

3.3.8 Libraries options

--with-libs='lib1 ...'

Link against additional libraries (default is no).

--with-system-expat

Build the `pyexpat` module using an installed `expat` library (default is no).

--with-system-libmpdec

Build the `_decimal` extension module using an installed `mpdec` library, see the `decimal` module (default is no).

Added in version 3.3.

--with-readline=editline

Use `editline` library for backend of the `readline` module.

Define the `WITH_EDITLINE` macro.

Added in version 3.10.

--without-readline

Don't build the `readline` module (built by default).

Don't define the `HAVE_LIBREADLINE` macro.

Added in version 3.10.

--with-libm=STRING

Override `libm` math library to *STRING* (default is system-dependent).

--with-libc=STRING

Override `libc` C library to *STRING* (default is system-dependent).

--with-openssl=DIR

Root of the OpenSSL directory.

Added in version 3.7.

--with-openssl-rpath=[no|auto|DIR]

Set runtime library directory (rpath) for OpenSSL libraries:

- `no` (default): don't set rpath;
- `auto`: auto-detect rpath from *`--with-openssl`* and `pkg-config`;
- *DIR*: set an explicit rpath.

Added in version 3.10.

3.3.9 Security Options

--with-hash-algorithm=[fnv|siphash13|siphash24]

Select hash algorithm for use in `Python/pyhash.c`:

- `siphash13` (default);
- `siphash24`;
- `fnv`.

Added in version 3.4.

Added in version 3.11: `siphash13` is added and it is the new default.

--with-builtin-hashlib-hashes=md5,sha1,sha256,sha512,sha3,blake2

Built-in hash modules:

- `md5`;
- `sha1`;
- `sha256`;
- `sha512`;
- `sha3` (with shake);
- `blake2`.

Added in version 3.9.

--with-ssl-default-suites=[python|openssl|STRING]

Override the OpenSSL default cipher suites string:

- `python` (default): use Python's preferred selection;
- `openssl`: leave OpenSSL's defaults untouched;
- *STRING*: use a custom string

See the `ssl` module.

Added in version 3.7.

Changed in version 3.10: The settings `python` and *STRING* also set TLS 1.2 as minimum protocol version.

3.3.10 macOS Options

See `Mac/README.rst`.

--enable-universalsdk

--enable-universalsdk=SDKDIR

Create a universal binary build. *SDKDIR* specifies which macOS SDK should be used to perform the build (default is no).

--enable-framework

--enable-framework=INSTALLDIR

Create a Python.framework rather than a traditional Unix install. Optional *INSTALLDIR* specifies the installation path (default is no).

--with-universal-archs=ARCH

Specify the kind of universal binary that should be created. This option is only valid when *--enable-universalsdk* is set.

Options:

- universal2;
- 32-bit;
- 64-bit;
- 3-way;
- intel;
- intel-32;
- intel-64;
- all.

--with-framework-name=FRAMEWORK

Specify the name for the python framework on macOS only valid when *--enable-framework* is set (default: Python).

3.3.11 Cross Compiling Options

Cross compiling, also known as cross building, can be used to build Python for another CPU architecture or platform. Cross compiling requires a Python interpreter for the build platform. The version of the build Python must match the version of the cross compiled host Python.

--build=BUILD

configure for building on BUILD, usually guessed by **config.guess**.

--host=HOST

cross-compile to build programs to run on HOST (target platform)

--with-build-python=path/to/python

path to build python binary for cross compiling

Added in version 3.11.

CONFIG_SITE=file

An environment variable that points to a file with configure overrides.

Example *config.site* file:

```
# config.site-aarch64
ac_cv_buggy_getaddrinfo=no
ac_cv_file__dev_ptmx=yes
ac_cv_file__dev_ptc=no
```

Cross compiling example:

```
CONFIG_SITE=config.site-aarch64 ../configure \
    --build=x86_64-pc-linux-gnu \
    --host=aarch64-unknown-linux-gnu \
    --with-build-python=../x86_64/python
```

3.4 Python Build System

3.4.1 Main files of the build system

- `configure.ac` => `configure`;
- `Makefile.pre.in` => `Makefile` (created by `configure`);
- `pyconfig.h` (created by `configure`);
- `Modules/Setup`: C extensions built by the Makefile using `Module/makesetup` shell script;

3.4.2 Main build steps

- C files (`.c`) are built as object files (`.o`).
- A static `libpython` library (`.a`) is created from objects files.
- `python.o` and the static `libpython` library are linked into the final `python` program.
- C extensions are built by the Makefile (see `Modules/Setup`).

3.4.3 Main Makefile targets

- `make`: Build Python with the standard library.
- `make platform:`: build the `python` program, but don't build the standard library extension modules.
- `make profile-opt`: build Python using Profile Guided Optimization (PGO). You can use the configure *`--enable-optimizations`* option to make this the default target of the `make` command (`make all` or just `make`).
- `make buildbottest`: Build Python and run the Python test suite, the same way than buildbots test Python. Set `TESTTIMEOUT` variable (in seconds) to change the test timeout (1200 by default: 20 minutes).
- `make install`: Build and install Python.
- `make regen-all`: Regenerate (almost) all generated files; `make regen-stdlib-module-names` and `autoconf` must be run separately for the remaining generated files.
- `make clean`: Remove built files.
- `make distclean`: Same than `make clean`, but remove also files created by the configure script.

3.4.4 C extensions

Some C extensions are built as built-in modules, like the `sys` module. They are built with the `Py_BUILD_CORE_BUILTIN` macro defined. Built-in modules have no `__file__` attribute:

```
>>> import sys
>>> sys
<module 'sys' (built-in)>
>>> sys.__file__
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
AttributeError: module 'sys' has no attribute '__file__'
```

Other C extensions are built as dynamic libraries, like the `_asyncio` module. They are built with the `Py_BUILD_CORE_MODULE` macro defined. Example on Linux x86-64:

```
>>> import _asyncio
>>> _asyncio
<module '_asyncio' from '/usr/lib64/python3.9/lib-dynload/_asyncio.cpython-39-x86_64-
↪linux-gnu.so'>
>>> _asyncio.__file__
'/usr/lib64/python3.9/lib-dynload/_asyncio.cpython-39-x86_64-linux-gnu.so'
```

`Modules/Setup` is used to generate Makefile targets to build C extensions. At the beginning of the files, C extensions are built as built-in modules. Extensions defined after the `*shared*` marker are built as dynamic libraries.

The `PyAPI_FUNC()`, `PyAPI_DATA()` and `PyMODINIT_FUNC` macros of `Include/exports.h` are defined differently depending if the `Py_BUILD_CORE_MODULE` macro is defined:

- Use `Py_EXPORTED_SYMBOL` if the `Py_BUILD_CORE_MODULE` is defined
- Use `Py_IMPORTED_SYMBOL` otherwise.

If the `Py_BUILD_CORE_BUILTIN` macro is used by mistake on a C extension built as a shared library, its `PyInit_xxx()` function is not exported, causing an `ImportError` on import.

3.5 Compiler and linker flags

Options set by the `./configure` script and environment variables and used by `Makefile`.

3.5.1 Preprocessor flags

CONFIGURE_CPPFLAGS

Value of *CPPFLAGS* variable passed to the `./configure` script.

Added in version 3.6.

CPPFLAGS

(Objective) C/C++ preprocessor flags, e.g. `-Iinclude_dir` if you have headers in a nonstandard directory *include_dir*.

Both *CPPFLAGS* and *LDFLAGS* need to contain the shell's value to be able to build extension modules using the directories specified in the environment variables.

BASECPPFLAGS

Added in version 3.4.

PY_CPPFLAGS

Extra preprocessor flags added for building the interpreter object files.

Default: `$(BASECPPFLAGS) -I. -I$(srcdir)/Include $(CONFIGURE_CPPFLAGS) $(CPPFLAGS)`.

Added in version 3.2.

3.5.2 Compiler flags

CC

C compiler command.

Example: `gcc -pthread`.

CXX

C++ compiler command.

Example: `g++ -pthread`.

CFLAGS

C compiler flags.

CFLAGS_NODIST

CFLAGS_NODIST is used for building the interpreter and stdlib C extensions. Use it when a compiler flag should *not* be part of *CFLAGS* once Python is installed (gh-65320).

In particular, *CFLAGS* should not contain:

- the compiler flag `-I` (for setting the search path for include files). The `-I` flags are processed from left to right, and any flags in *CFLAGS* would take precedence over user- and package-supplied `-I` flags.
- hardening flags such as `-Werror` because distributions cannot control whether packages installed by users conform to such heightened standards.

Added in version 3.5.

COMPILEALL_OPTS

Options passed to the `compileall` command line when building PYC files in `make install`. Default: `-j0`.

Added in version 3.12.

EXTRA_CFLAGS

Extra C compiler flags.

CONFIGURE_CFLAGS

Value of *CFLAGS* variable passed to the `./configure` script.

Added in version 3.2.

CONFIGURE_CFLAGS_NODIST

Value of *CFLAGS_NODIST* variable passed to the `./configure` script.

Added in version 3.5.

BASECFLAGS

Base compiler flags.

OPT

Optimization flags.

CFLAGS_ALIASING

Strict or non-strict aliasing flags used to compile `Python/dtoa.c`.

Added in version 3.7.

CCSHARED

Compiler flags used to build a shared library.

For example, `-fPIC` is used on Linux and on BSD.

CFLAGSFORSHARED

Extra C flags added for building the interpreter object files.

Default: `$(CCSHARED)` when `--enable-shared` is used, or an empty string otherwise.

PY_CFLAGS

Default: `$(BASECFLAGS) $(OPT) $(CONFIGURE_CFLAGS) $(CFLAGS) $(EXTRA_CFLAGS)`.

PY_CFLAGS_NODIST

Default: `$(CONFIGURE_CFLAGS_NODIST) $(CFLAGS_NODIST) -I$(srcdir)/Include/internal`.

Added in version 3.5.

PY_STDMODULE_CFLAGS

C flags used for building the interpreter object files.

Default: `$(PY_CFLAGS) $(PY_CFLAGS_NODIST) $(PY_CPPFLAGS) $(CFLAGSFORSHARED)`.

Added in version 3.7.

PY_CORE_CFLAGS

Default: `$(PY_STDMODULE_CFLAGS) -DPy_BUILD_CORE`.

Added in version 3.2.

PY_BUILTIN_MODULE_CFLAGS

Compiler flags to build a standard library extension module as a built-in module, like the `posix` module.

Default: `$(PY_STDMODULE_CFLAGS) -DPy_BUILD_CORE_BUILTIN`.

Added in version 3.8.

PURIFY

Purify command. Purify is a memory debugger program.

Default: empty string (not used).

3.5.3 Linker flags

LINKCC

Linker command used to build programs like `python` and `_testembed`.

Default: `$(PURIFY) $(CC)`.

CONFIGURE_LDFLAGS

Value of *LDFLAGS* variable passed to the `./configure` script.

Avoid assigning *CFLAGS*, *LDFLAGS*, etc. so users can use them on the command line to append to these values without stomping the pre-set values.

Added in version 3.2.

LDFLAGS_NODIST

LDFLAGS_NODIST is used in the same manner as *CFLAGS_NODIST*. Use it when a linker flag should *not* be part of *LDFLAGS* once Python is installed (gh-65320).

In particular, *LDFLAGS* should not contain:

- the compiler flag `-L` (for setting the search path for libraries). The `-L` flags are processed from left to right, and any flags in *LDFLAGS* would take precedence over user- and package-supplied `-L` flags.

CONFIGURE_LDFLAGS_NODIST

Value of *LDFLAGS_NODIST* variable passed to the `./configure` script.

Added in version 3.8.

LDFLAGS

Linker flags, e.g. `-Llib_dir` if you have libraries in a nonstandard directory *lib_dir*.

Both *CPPFLAGS* and *LDFLAGS* need to contain the shell's value to be able to build extension modules using the directories specified in the environment variables.

LIBS

Linker flags to pass libraries to the linker when linking the Python executable.

Example: `-lrt`.

LDSHARED

Command to build a shared library.

Default: `@LDSHARED@ $(PY_LDFLAGS)`.

BLDSHARED

Command to build `libpython` shared library.

Default: `@BLDSHARED@ $(PY_CORE_LDFLAGS)`.

PY_LDFLAGS

Default: `$(CONFIGURE_LDFLAGS) $(LDFLAGS)`.

PY_LDFLAGS_NODIST

Default: `$(CONFIGURE_LDFLAGS_NODIST) $(LDFLAGS_NODIST)`.

Added in version 3.8.

PY_CORE_LDFLAGS

Linker flags used for building the interpreter object files.

Added in version 3.8.

CHAPTER

FOUR

USING PYTHON ON WINDOWS

This document aims to give an overview of Windows-specific behaviour you should know about when using Python on Microsoft Windows.

Unlike most Unix systems and services, Windows does not include a system supported installation of Python. To make Python available, the CPython team has compiled Windows installers with every release for many years. These installers are primarily intended to add a per-user installation of Python, with the core interpreter and library being used by a single user. The installer is also able to install for all users of a single machine, and a separate ZIP file is available for application-local distributions.

As specified in **PEP 11**, a Python release only supports a Windows platform while Microsoft considers the platform under extended support. This means that Python 3.12 supports Windows 8.1 and newer. If you require Windows 7 support, please install Python 3.8.

There are a number of different installers available for Windows, each with certain benefits and downsides.

The full installer contains all components and is the best option for developers using Python for any kind of project.

The Microsoft Store package is a simple installation of Python that is suitable for running scripts and packages, and using IDLE or other development environments. It requires Windows 10 and above, but can be safely installed without corrupting other programs. It also provides many convenient commands for launching Python and its tools.

The nuget.org packages are lightweight installations intended for continuous integration systems. It can be used to build Python packages or run scripts, but is not updateable and has no user interface tools.

The embeddable package is a minimal package of Python suitable for embedding into a larger application.

4.1 The full installer

4.1.1 Installation steps

Four Python 3.12 installers are available for download - two each for the 32-bit and 64-bit versions of the interpreter. The *web installer* is a small initial download, and it will automatically download the required components as necessary. The *offline installer* includes the components necessary for a default installation and only requires an internet connection for optional features. See *Installing Without Downloading* for other ways to avoid downloading during installation.

After starting the installer, one of two options may be selected:

If you select "Install Now":

- You will *not* need to be an administrator (unless a system update for the C Runtime Library is required or you install the *Python Launcher for Windows* for all users)
- Python will be installed into your user directory
- The *Python Launcher for Windows* will be installed according to the option at the bottom of the first page
- The standard library, test suite, launcher and pip will be installed
- If selected, the install directory will be added to your `PATH`
- Shortcuts will only be visible for the current user

Selecting "Customize installation" will allow you to select the features to install, the installation location and other options or post-install actions. To install debugging symbols or binaries, you will need to use this option.

To perform an all-users installation, you should select "Customize installation". In this case:

- You may be required to provide administrative credentials or approval
- Python will be installed into the Program Files directory
- The *Python Launcher for Windows* will be installed into the Windows directory
- Optional features may be selected during installation
- The standard library can be pre-compiled to bytecode
- If selected, the install directory will be added to the system `PATH`
- Shortcuts are available for all users

4.1.2 Removing the MAX_PATH Limitation

Windows historically has limited path lengths to 260 characters. This meant that paths longer than this would not resolve and errors would result.

In the latest versions of Windows, this limitation can be expanded to approximately 32,000 characters. Your administrator will need to activate the "Enable Win32 long paths" group policy, or set `LongPathsEnabled` to `1` in the registry key `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem`.

This allows the `open()` function, the `os` module and most other path functionality to accept and return paths longer than 260 characters.

After changing the above option, no further configuration is required.

Changed in version 3.6: Support for long paths was enabled in Python.

4.1.3 Installing Without UI

All of the options available in the installer UI can also be specified from the command line, allowing scripted installers to replicate an installation on many machines without user interaction. These options may also be set without suppressing the UI in order to change some of the defaults.

The following options (found by executing the installer with `/?`) can be passed into the installer:

Name	Description
/passive	to display progress without requiring user interaction
/quiet	to install/uninstall without displaying any UI
/simple	to prevent user customization
/uninstall	to remove Python (without confirmation)
/layout [directory]	to pre-download all components
/log [filename]	to specify log files location

All other options are passed as `name=value`, where the value is usually `0` to disable a feature, `1` to enable a feature, or a path. The full list of available options is shown below.

Name	Description	Default
InstallAllUsers	Perform a system-wide installation.	0
TargetDir	The installation directory	Selected based on InstallAllUsers
DefaultAllUsersTargetDir	The default installation directory for all-user installs	`%ProgramFiles%\Python X.Y` or `%ProgramFiles(x86)%\Python X.Y`
DefaultJustForMeTargetDir	The default install directory for just-for-me installs	`%LocalAppData%\Programs\Python\PythonXY` or `%LocalAppData%\Programs\Python\PythonXY-32` or `%LocalAppData%\Programs\Python\PythonXY-64`
DefaultCustomTargetDir	The default custom install directory displayed in the UI	(empty)
AssociateFiles	Create file associations if the launcher is also installed.	1
CompileAll	Compile all `.py` files to `.pyc`.	0
PrependPath	Prepend install and Scripts directories to `PATH` and add `.PY` to `PATHEXT`	0
AppendPath	Append install and Scripts directories to `PATH` and add `.PY` to `PATHEXT`	0
Shortcuts	Create shortcuts for the interpreter, documentation and IDLE if installed.	1
Include_doc	Install Python manual	1
Include_debu	Install debug binaries	0
Include_dev	Install developer headers and libraries. Omitting this may lead to an unusable installation.	1
Include_exe	Install `python.exe` and related files. Omitting this may lead to an unusable installation.	1
Include_laun	Install *Python Launcher for Windows*.	1
InstallLauncherAllUsers	Installs the launcher for all users. Also requires `Include_launcher` to be set to 1	1
Include_lib	Install standard library and extension modules. Omitting this may lead to an unusable installation.	1
Include_pip	Install bundled pip and setuptools	1
Include_symb	Install debugging symbols (`*.pdb`)	0
Include_tcltk	Install Tcl/Tk support and IDLE	1
Include_test	Install standard library test suite	1
Include_tools	Install utility scripts	1

For example, to silently install a default, system-wide Python installation, you could use the following command (from an elevated command prompt):

```
python-3.9.0.exe /quiet InstallAllUsers=1 PrependPath=1 Include_test=0
```

To allow users to easily install a personal copy of Python without the test suite, you could provide a shortcut with the following command. This will display a simplified initial page and disallow customization:

```
python-3.9.0.exe InstallAllUsers=0 Include_launcher=0 Include_test=0
    SimpleInstall=1 SimpleInstallDescription="Just for me, no test suite."
```

(Note that omitting the launcher also omits file associations, and is only recommended for per-user installs when there is also a system-wide installation that included the launcher.)

The options listed above can also be provided in a file named `unattend.xml` alongside the executable. This file specifies a list of options and values. When a value is provided as an attribute, it will be converted to a number if possible. Values provided as element text are always left as strings. This example file sets the same options as the previous example:

```
<Options>
    <Option Name="InstallAllUsers" Value="no" />
    <Option Name="Include_launcher" Value="0" />
    <Option Name="Include_test" Value="no" />
    <Option Name="SimpleInstall" Value="yes" />
    <Option Name="SimpleInstallDescription">Just for me, no test suite</Option>
</Options>
```

4.1.4 Installing Without Downloading

As some features of Python are not included in the initial installer download, selecting those features may require an internet connection. To avoid this need, all possible components may be downloaded on-demand to create a complete *layout* that will no longer require an internet connection regardless of the selected features. Note that this download may be bigger than required, but where a large number of installations are going to be performed it is very useful to have a locally cached copy.

Execute the following command from Command Prompt to download all possible required files. Remember to substitute `python-3.9.0.exe` for the actual name of your installer, and to create layouts in their own directories to avoid collisions between files with the same name.

```
python-3.9.0.exe /layout [optional target directory]
```

You may also specify the `/quiet` option to hide the progress display.

4.1.5 Modifying an install

Once Python has been installed, you can add or remove features through the Programs and Features tool that is part of Windows. Select the Python entry and choose "Uninstall/Change" to open the installer in maintenance mode.

"Modify" allows you to add or remove features by modifying the checkboxes - unchanged checkboxes will not install or remove anything. Some options cannot be changed in this mode, such as the install directory; to modify these, you will need to remove and then reinstall Python completely.

"Repair" will verify all the files that should be installed using the current settings and replace any that have been removed or modified.

"Uninstall" will remove Python entirely, with the exception of the *Python Launcher for Windows*, which has its own entry in Programs and Features.

4.2 The Microsoft Store package

Added in version 3.7.2.

The Microsoft Store package is an easily installable Python interpreter that is intended mainly for interactive use, for example, by students.

To install the package, ensure you have the latest Windows 10 updates and search the Microsoft Store app for "Python 3.12". Ensure that the app you select is published by the Python Software Foundation, and install it.

> **Warning:** Python will always be available for free on the Microsoft Store. If you are asked to pay for it, you have not selected the correct package.

After installation, Python may be launched by finding it in Start. Alternatively, it will be available from any Command Prompt or PowerShell session by typing `python`. Further, pip and IDLE may be used by typing `pip` or `idle`. IDLE can also be found in Start.

All three commands are also available with version number suffixes, for example, as `python3.exe` and `python3.x.exe` as well as `python.exe` (where `3.x` is the specific version you want to launch, such as 3.12). Open "Manage App Execution Aliases" through Start to select which version of Python is associated with each command. It is recommended to make sure that `pip` and `idle` are consistent with whichever version of `python` is selected.

Virtual environments can be created with `python -m venv` and activated and used as normal.

If you have installed another version of Python and added it to your `PATH` variable, it will be available as `python.exe` rather than the one from the Microsoft Store. To access the new installation, use `python3.exe` or `python3.x.exe`.

The `py.exe` launcher will detect this Python installation, but will prefer installations from the traditional installer.

To remove Python, open Settings and use Apps and Features, or else find Python in Start and right-click to select Uninstall. Uninstalling will remove all packages you installed directly into this Python installation, but will not remove any virtual environments

4.2.1 Known issues

Redirection of local data, registry, and temporary paths

Because of restrictions on Microsoft Store apps, Python scripts may not have full write access to shared locations such as `TEMP` and the registry. Instead, it will write to a private copy. If your scripts must modify the shared locations, you will need to install the full installer.

At runtime, Python will use a private copy of well-known Windows folders and the registry. For example, if the environment variable `%APPDATA%` is `c:\Users\<user>\AppData\`, then when writing to `C:\Users\<user>\AppData\Local` will write to `C:\Users\<user>\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.8_qbz5n2kfra8p0\LocalCache\Local\`.

When reading files, Windows will return the file from the private folder, or if that does not exist, the real Windows directory. For example reading `C:\Windows\System32` returns the contents of `C:\Windows\System32` plus the contents of `C:\Program Files\WindowsApps\package_name\VFS\SystemX86`.

You can find the real path of any existing file using `os.path.realpath()`:

```
>>> import os
>>> test_file = 'C:\\Users\\example\\AppData\\Local\\test.txt'
>>> os.path.realpath(test_file)
```

(continues on next page)

(continued from previous page)

```
'C:\\Users\\example\\AppData\\Local\\Packages\\PythonSoftwareFoundation.Python.3.8_
↪qbz5n2kfra8p0\\LocalCache\\Local\\test.txt'
```

When writing to the Windows Registry, the following behaviors exist:

- Reading from `HKLM\\Software` is allowed and results are merged with the `registry.dat` file in the package.
- Writing to `HKLM\\Software` is not allowed if the corresponding key/value exists, i.e. modifying existing keys.
- Writing to `HKLM\\Software` is allowed as long as a corresponding key/value does not exist in the package and the user has the correct access permissions.

For more detail on the technical basis for these limitations, please consult Microsoft's documentation on packaged full-trust apps, currently available at docs.microsoft.com/en-us/windows/msix/desktop/desktop-to-uwp-behind-the-scenes

4.3 The nuget.org packages

Added in version 3.5.2.

The nuget.org package is a reduced size Python environment intended for use on continuous integration and build systems that do not have a system-wide install of Python. While nuget is "the package manager for .NET", it also works perfectly fine for packages containing build-time tools.

Visit nuget.org for the most up-to-date information on using nuget. What follows is a summary that is sufficient for Python developers.

The `nuget.exe` command line tool may be downloaded directly from `https://aka.ms/nugetclidl`, for example, using curl or PowerShell. With the tool, the latest version of Python for 64-bit or 32-bit machines is installed using:

```
nuget.exe install python -ExcludeVersion -OutputDirectory .
nuget.exe install pythonx86 -ExcludeVersion -OutputDirectory .
```

To select a particular version, add a `-Version 3.x.y`. The output directory may be changed from `.`, and the package will be installed into a subdirectory. By default, the subdirectory is named the same as the package, and without the `-ExcludeVersion` option this name will include the specific version installed. Inside the subdirectory is a `tools` directory that contains the Python installation:

```
# Without -ExcludeVersion
> .\python.3.5.2\tools\python.exe -V
Python 3.5.2

# With -ExcludeVersion
> .\python\tools\python.exe -V
Python 3.5.2
```

In general, nuget packages are not upgradeable, and newer versions should be installed side-by-side and referenced using the full path. Alternatively, delete the package directory manually and install it again. Many CI systems will do this automatically if they do not preserve files between builds.

Alongside the `tools` directory is a `build\native` directory. This contains a MSBuild properties file `python.props` that can be used in a C++ project to reference the Python install. Including the settings will automatically use the headers and import libraries in your build.

The package information pages on nuget.org are www.nuget.org/packages/python for the 64-bit version and www.nuget.org/packages/pythonx86 for the 32-bit version.

4.4 The embeddable package

Added in version 3.5.

The embedded distribution is a ZIP file containing a minimal Python environment. It is intended for acting as part of another application, rather than being directly accessed by end-users.

When extracted, the embedded distribution is (almost) fully isolated from the user's system, including environment variables, system registry settings, and installed packages. The standard library is included as pre-compiled and optimized `.pyc` files in a ZIP, and `python3.dll`, `python37.dll`, `python.exe` and `pythonw.exe` are all provided. Tcl/tk (including all dependents, such as Idle), pip and the Python documentation are not included.

Note: The embedded distribution does not include the Microsoft C Runtime and it is the responsibility of the application installer to provide this. The runtime may have already been installed on a user's system previously or automatically via Windows Update, and can be detected by finding `ucrtbase.dll` in the system directory.

Third-party packages should be installed by the application installer alongside the embedded distribution. Using pip to manage dependencies as for a regular Python installation is not supported with this distribution, though with some care it may be possible to include and use pip for automatic updates. In general, third-party packages should be treated as part of the application ("vendoring") so that the developer can ensure compatibility with newer versions before providing updates to users.

The two recommended use cases for this distribution are described below.

4.4.1 Python Application

An application written in Python does not necessarily require users to be aware of that fact. The embedded distribution may be used in this case to include a private version of Python in an install package. Depending on how transparent it should be (or conversely, how professional it should appear), there are two options.

Using a specialized executable as a launcher requires some coding, but provides the most transparent experience for users. With a customized launcher, there are no obvious indications that the program is running on Python: icons can be customized, company and version information can be specified, and file associations behave properly. In most cases, a custom launcher should simply be able to call `Py_Main` with a hard-coded command line.

The simpler approach is to provide a batch file or generated shortcut that directly calls the `python.exe` or `pythonw.exe` with the required command-line arguments. In this case, the application will appear to be Python and not its actual name, and users may have trouble distinguishing it from other running Python processes or file associations.

With the latter approach, packages should be installed as directories alongside the Python executable to ensure they are available on the path. With the specialized launcher, packages can be located in other locations as there is an opportunity to specify the search path before launching the application.

4.4.2 Embedding Python

Applications written in native code often require some form of scripting language, and the embedded Python distribution can be used for this purpose. In general, the majority of the application is in native code, and some part will either invoke `python.exe` or directly use `python3.dll`. For either case, extracting the embedded distribution to a subdirectory of the application installation is sufficient to provide a loadable Python interpreter.

As with the application use, packages can be installed to any location as there is an opportunity to specify search paths before initializing the interpreter. Otherwise, there is no fundamental differences between using the embedded distribution and a regular installation.

4.5 Alternative bundles

Besides the standard CPython distribution, there are modified packages including additional functionality. The following is a list of popular versions and their key features:

ActivePython
Installer with multi-platform compatibility, documentation, PyWin32

Anaconda
Popular scientific modules (such as numpy, scipy and pandas) and the `conda` package manager.

Enthought Deployment Manager
"The Next Generation Python Environment and Package Manager".

Previously Enthought provided Canopy, but it reached end of life in 2016.

WinPython
Windows-specific distribution with prebuilt scientific packages and tools for building packages.

Note that these packages may not include the latest versions of Python or other libraries, and are not maintained or supported by the core Python team.

4.6 Configuring Python

To run Python conveniently from a command prompt, you might consider changing some default environment variables in Windows. While the installer provides an option to configure the PATH and PATHEXT variables for you, this is only reliable for a single, system-wide installation. If you regularly use multiple versions of Python, consider using the *Python Launcher for Windows*.

4.6.1 Excursus: Setting environment variables

Windows allows environment variables to be configured permanently at both the User level and the System level, or temporarily in a command prompt.

To temporarily set environment variables, open Command Prompt and use the **`set`** command:

```
C:\>set PATH=C:\Program Files\Python 3.9;%PATH%
C:\>set PYTHONPATH=%PYTHONPATH%;C:\My_python_lib
C:\>python
```

These changes will apply to any further commands executed in that console, and will be inherited by any applications started from the console.

Including the variable name within percent signs will expand to the existing value, allowing you to add your new value at either the start or the end. Modifying `PATH` by adding the directory containing **`python.exe`** to the start is a common way to ensure the correct version of Python is launched.

To permanently modify the default environment variables, click Start and search for 'edit environment variables', or open System properties, *Advanced system settings* and click the *Environment Variables* button. In this dialog, you can add or modify User and System variables. To change System variables, you need non-restricted access to your machine (i.e. Administrator rights).

Note: Windows will concatenate User variables *after* System variables, which may cause unexpected results when modifying `PATH`.

The *PYTHONPATH* variable is used by all versions of Python, so you should not permanently configure it unless the listed paths only include code that is compatible with all of your installed Python versions.

See also:

https://docs.microsoft.com/en-us/windows/win32/procthread/environment-variables
Overview of environment variables on Windows

https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/set_1
The `set` command, for temporarily modifying environment variables

https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/setx
The `setx` command, for permanently modifying environment variables

4.6.2 Finding the Python executable

Changed in version 3.5.

Besides using the automatically created start menu entry for the Python interpreter, you might want to start Python in the command prompt. The installer has an option to set that up for you.

On the first page of the installer, an option labelled "Add Python to PATH" may be selected to have the installer add the install location into the `PATH`. The location of the `Scripts\` folder is also added. This allows you to type **python** to run the interpreter, and **pip** for the package installer. Thus, you can also execute your scripts with command line options, see *Command line* documentation.

If you don't enable this option at install time, you can always re-run the installer, select Modify, and enable it. Alternatively, you can manually modify the `PATH` using the directions in *Excursus: Setting environment variables*. You need to set your `PATH` environment variable to include the directory of your Python installation, delimited by a semicolon from other entries. An example variable could look like this (assuming the first two entries already existed):

```
C:\WINDOWS\system32;C:\WINDOWS;C:\Program Files\Python 3.9
```

4.7 UTF-8 mode

Added in version 3.7.

Windows still uses legacy encodings for the system encoding (the ANSI Code Page). Python uses it for the default encoding of text files (e.g. `locale.getencoding()`).

This may cause issues because UTF-8 is widely used on the internet and most Unix systems, including WSL (Windows Subsystem for Linux).

You can use the Python UTF-8 Mode to change the default text encoding to UTF-8. You can enable the Python UTF-8 Mode via the `-X utf8` command line option, or the `PYTHONUTF8=1` environment variable. See *PYTHONUTF8* for enabling UTF-8 mode, and *Excursus: Setting environment variables* for how to modify environment variables.

When the Python UTF-8 Mode is enabled, you can still use the system encoding (the ANSI Code Page) via the "mbcs" codec.

Note that adding `PYTHONUTF8=1` to the default environment variables will affect all Python 3.7+ applications on your system. If you have any Python 3.7+ applications which rely on the legacy system encoding, it is recommended to set the environment variable temporarily or use the `-X utf8` command line option.

Note: Even when UTF-8 mode is disabled, Python uses UTF-8 by default on Windows for:

- Console I/O including standard I/O (see **PEP 528** for details).
- The *filesystem encoding* (see **PEP 529** for details).

4.8 Python Launcher for Windows

Added in version 3.3.

The Python launcher for Windows is a utility which aids in locating and executing of different Python versions. It allows scripts (or the command-line) to indicate a preference for a specific Python version, and will locate and execute that version.

Unlike the `PATH` variable, the launcher will correctly select the most appropriate version of Python. It will prefer per-user installations over system-wide ones, and orders by language version rather than using the most recently installed version.

The launcher was originally specified in **PEP 397**.

4.8.1 Getting started

From the command-line

Changed in version 3.6.

System-wide installations of Python 3.3 and later will put the launcher on your `PATH`. The launcher is compatible with all available versions of Python, so it does not matter which version is installed. To check that the launcher is available, execute the following command in Command Prompt:

```
py
```

You should find that the latest version of Python you have installed is started - it can be exited as normal, and any additional command-line arguments specified will be sent directly to Python.

If you have multiple versions of Python installed (e.g., 3.7 and 3.12) you will have noticed that Python 3.12 was started - to launch Python 3.7, try the command:

```
py -3.7
```

If you want the latest version of Python 2 you have installed, try the command:

```
py -2
```

If you see the following error, you do not have the launcher installed:

```
'py' is not recognized as an internal or external command,
operable program or batch file.
```

The command:

```
py --list
```

displays the currently installed version(s) of Python.

The `-x.y` argument is the short form of the `-V:Company/Tag` argument, which allows selecting a specific Python runtime, including those that may have come from somewhere other than python.org. Any runtime registered by following **PEP 514** will be discoverable. The `--list` command lists all available runtimes using the `-V:` format.

When using the `-V:` argument, specifying the Company will limit selection to runtimes from that provider, while specifying only the Tag will select from all providers. Note that omitting the slash implies a tag:

```
# Select any '3.*' tagged runtime
py -V:3

# Select any 'PythonCore' released runtime
py -V:PythonCore/

# Select PythonCore's latest Python 3 runtime
py -V:PythonCore/3
```

The short form of the argument (`-3`) only ever selects from core Python releases, and not other distributions. However, the longer form (`-V:3`) will select from any.

The Company is matched on the full string, case-insenitive. The Tag is matched oneither the full string, or a prefix, provided the next character is a dot or a hyphen. This allows `-V:3.1` to match `3.1-32`, but not `3.10`. Tags are sorted using numerical ordering (`3.10` is newer than `3.1`), but are compared using text (`-V:3.01` does not match `3.1`).

Virtual environments

Added in version 3.5.

If the launcher is run with no explicit Python version specification, and a virtual environment (created with the standard library `venv` module or the external `virtualenv` tool) active, the launcher will run the virtual environment's interpreter rather than the global one. To run the global interpreter, either deactivate the virtual environment, or explicitly specify the global Python version.

From a script

Let's create a test Python script - create a file called `hello.py` with the following contents

```
#! python
import sys
sys.stdout.write("hello from Python %s\n" % (sys.version,))
```

From the directory in which hello.py lives, execute the command:

```
py hello.py
```

You should notice the version number of your latest Python 2.x installation is printed. Now try changing the first line to be:

```
#! python3
```

Re-executing the command should now print the latest Python 3.x information. As with the above command-line examples, you can specify a more explicit version qualifier. Assuming you have Python 3.7 installed, try changing the first line to `#! python3.7` and you should find the 3.7 version information printed.

Note that unlike interactive use, a bare "python" will use the latest version of Python 2.x that you have installed. This is for backward compatibility and for compatibility with Unix, where the command `python` typically refers to Python 2.

From file associations

The launcher should have been associated with Python files (i.e. `.py`, `.pyw`, `.pyc` files) when it was installed. This means that when you double-click on one of these files from Windows explorer the launcher will be used, and therefore you can use the same facilities described above to have the script specify the version which should be used.

The key benefit of this is that a single launcher can support multiple Python versions at the same time depending on the contents of the first line.

4.8.2 Shebang Lines

If the first line of a script file starts with `#!`, it is known as a "shebang" line. Linux and other Unix like operating systems have native support for such lines and they are commonly used on such systems to indicate how a script should be executed. This launcher allows the same facilities to be used with Python scripts on Windows and the examples above demonstrate their use.

To allow shebang lines in Python scripts to be portable between Unix and Windows, this launcher supports a number of 'virtual' commands to specify which interpreter to use. The supported virtual commands are:

- `/usr/bin/env`
- `/usr/bin/python`
- `/usr/local/bin/python`
- `python`

For example, if the first line of your script starts with

```
#! /usr/bin/python
```

The default Python will be located and used. As many Python scripts written to work on Unix will already have this line, you should find these scripts can be used by the launcher without modification. If you are writing a new script on Windows which you hope will be useful on Unix, you should use one of the shebang lines starting with `/usr`.

Any of the above virtual commands can be suffixed with an explicit version (either just the major version, or the major and minor version). Furthermore the 32-bit version can be requested by adding "-32" after the minor version. I.e. `/usr/bin/python3.7-32` will request usage of the 32-bit python 3.7.

Added in version 3.7: Beginning with python launcher 3.7 it is possible to request 64-bit version by the "-64" suffix. Furthermore it is possible to specify a major and architecture without minor (i.e. `/usr/bin/python3-64`).

Changed in version 3.11: The "-64" suffix is deprecated, and now implies "any architecture that is not provably i386/32-bit". To request a specific environment, use the new `-V:`*`TAG`* argument with the complete tag.

The `/usr/bin/env` form of shebang line has one further special property. Before looking for installed Python interpreters, this form will search the executable `PATH` for a Python executable matching the name provided as the first argument. This corresponds to the behaviour of the Unix `env` program, which performs a `PATH` search. If an executable matching the first argument after the `env` command cannot be found, but the argument starts with `python`, it will be handled as described for the other virtual commands. The environment variable `PYLAUNCHER_NO_SEARCH_PATH` may be set (to any value) to skip this search of `PATH`.

Shebang lines that do not match any of these patterns are looked up in the `[commands]` section of the launcher's *.INI file*. This may be used to handle certain commands in a way that makes sense for your system. The name of the command must be a single argument (no spaces in the shebang executable), and the value substituted is the full path to the executable (additional arguments specified in the .INI will be quoted as part of the filename).

```
[commands]
/bin/xpython=C:\Program Files\XPython\python.exe
```

Any commands not found in the .INI file are treated as **Windows** executable paths that are absolute or relative to the directory containing the script file. This is a convenience for Windows-only scripts, such as those generated by an installer, since the behavior is not compatible with Unix-style shells. These paths may be quoted, and may include multiple arguments, after which the path to the script and any additional arguments will be appended.

4.8.3 Arguments in shebang lines

The shebang lines can also specify additional options to be passed to the Python interpreter. For example, if you have a shebang line:

```
#! /usr/bin/python -v
```

Then Python will be started with the `-v` option

4.8.4 Customization

Customization via INI files

Two .ini files will be searched by the launcher - `py.ini` in the current user's application data directory (`%LOCALAPPDATA%` or `$env:LocalAppData`) and `py.ini` in the same directory as the launcher. The same .ini files are used for both the 'console' version of the launcher (i.e. py.exe) and for the 'windows' version (i.e. pyw.exe).

Customization specified in the "application directory" will have precedence over the one next to the executable, so a user, who may not have write access to the .ini file next to the launcher, can override commands in that global .ini file.

Customizing default Python versions

In some cases, a version qualifier can be included in a command to dictate which version of Python will be used by the command. A version qualifier starts with a major version number and can optionally be followed by a period ('.') and a minor version specifier. Furthermore it is possible to specify if a 32 or 64 bit implementation shall be requested by adding "-32" or "-64".

For example, a shebang line of `#!python` has no version qualifier, while `#!python3` has a version qualifier which specifies only a major version.

If no version qualifiers are found in a command, the environment variable `PY_PYTHON` can be set to specify the default version qualifier. If it is not set, the default is "3". The variable can specify any value that may be passed on the command line, such as "3", "3.7", "3.7-32" or "3.7-64". (Note that the "-64" option is only available with the launcher included with Python 3.7 or newer.)

If no minor version qualifiers are found, the environment variable `PY_PYTHON{major}` (where `{major}` is the current major version qualifier as determined above) can be set to specify the full version. If no such option is found, the launcher will enumerate the installed Python versions and use the latest minor release found for the major version, which is likely, although not guaranteed, to be the most recently installed version in that family.

On 64-bit Windows with both 32-bit and 64-bit implementations of the same (major.minor) Python version installed, the 64-bit version will always be preferred. This will be true for both 32-bit and 64-bit implementations of the launcher - a 32-bit launcher will prefer to execute a 64-bit Python installation of the specified version if available. This is so the behavior of the launcher can be predicted knowing only what versions are installed on the PC and without regard to the order in which they were installed (i.e., without knowing whether a 32 or 64-bit version of Python and corresponding launcher was installed last). As noted above, an optional "-32" or "-64" suffix can be used on a version specifier to change this behaviour.

Examples:

- If no relevant options are set, the commands `python` and `python2` will use the latest Python 2.x version installed and the command `python3` will use the latest Python 3.x installed.
- The command `python3.7` will not consult any options at all as the versions are fully specified.
- If `PY_PYTHON=3`, the commands `python` and `python3` will both use the latest installed Python 3 version.
- If `PY_PYTHON=3.7-32`, the command `python` will use the 32-bit implementation of 3.7 whereas the command `python3` will use the latest installed Python (PY_PYTHON was not considered at all as a major version was specified.)
- If `PY_PYTHON=3` and `PY_PYTHON3=3.7`, the commands `python` and `python3` will both use specifically 3.7

In addition to environment variables, the same settings can be configured in the .INI file used by the launcher. The section in the INI file is called `[defaults]` and the key name will be the same as the environment variables without the leading `PY_` prefix (and note that the key names in the INI file are case insensitive.) The contents of an environment variable will override things specified in the INI file.

For example:

- Setting `PY_PYTHON=3.7` is equivalent to the INI file containing:

```
[defaults]
python=3.7
```

- Setting `PY_PYTHON=3` and `PY_PYTHON3=3.7` is equivalent to the INI file containing:

```
[defaults]
python=3
python3=3.7
```

4.8.5 Diagnostics

If an environment variable `PYLAUNCHER_DEBUG` is set (to any value), the launcher will print diagnostic information to stderr (i.e. to the console). While this information manages to be simultaneously verbose *and* terse, it should allow you to see what versions of Python were located, why a particular version was chosen and the exact command-line used to execute the target Python. It is primarily intended for testing and debugging.

4.8.6 Dry Run

If an environment variable `PYLAUNCHER_DRYRUN` is set (to any value), the launcher will output the command it would have run, but will not actually launch Python. This may be useful for tools that want to use the launcher to detect and then launch Python directly. Note that the command written to standard output is always encoded using UTF-8, and may not render correctly in the console.

4.8.7 Install on demand

If an environment variable `PYLAUNCHER_ALLOW_INSTALL` is set (to any value), and the requested Python version is not installed but is available on the Microsoft Store, the launcher will attempt to install it. This may require user interaction to complete, and you may need to run the command again.

An additional `PYLAUNCHER_ALWAYS_INSTALL` variable causes the launcher to always try to install Python, even if it is detected. This is mainly intended for testing (and should be used with `PYLAUNCHER_DRYRUN`).

4.8.8 Return codes

The following exit codes may be returned by the Python launcher. Unfortunately, there is no way to distinguish these from the exit code of Python itself.

The names of codes are as used in the sources, and are only for reference. There is no way to access or resolve them apart from reading this page. Entries are listed in alphabetical order of names.

Name	Value	Description
RC_BAD_VENV_CFG	107	A `pyvenv.cfg` was found but is corrupt.
RC_CREATE_PROCESS	101	Failed to launch Python.
RC_INSTALLING	111	An install was started, but the command will need to be re-run after it completes.
RC_INTERNAL_ERROR	109	Unexpected error. Please report a bug.
RC_NO_COMMANDLINE	108	Unable to obtain command line from the operating system.
RC_NO_PYTHON	103	Unable to locate the requested version.
RC_NO_VENV_CFG	106	A `pyvenv.cfg` was required but not found.

4.9 Finding modules

These notes supplement the description at sys-path-init with detailed Windows notes.

When no `._pth` file is found, this is how `sys.path` is populated on Windows:

- An empty entry is added at the start, which corresponds to the current directory.
- If the environment variable *`PYTHONPATH`* exists, as described in *Environment variables*, its entries are added next. Note that on Windows, paths in this variable must be separated by semicolons, to distinguish them from the colon used in drive identifiers (`C:\` etc.).
- Additional "application paths" can be added in the registry as subkeys of `\SOFTWARE\Python\PythonCore{version}\PythonPath` under both the `HKEY_CURRENT_USER` and `HKEY_LOCAL_MACHINE` hives. Subkeys which have semicolon-delimited path strings as their default value will cause each path to be added to `sys.path`. (Note that all known installers only use HKLM, so HKCU is typically empty.)
- If the environment variable *`PYTHONHOME`* is set, it is assumed as "Python Home". Otherwise, the path of the main Python executable is used to locate a "landmark file" (either `Lib\os.py` or `pythonXY.zip`) to deduce the "Python Home". If a Python home is found, the relevant sub-directories added to `sys.path` (`Lib`, `plat-win`, etc) are based on that folder. Otherwise, the core Python path is constructed from the PythonPath stored in the registry.
- If the Python Home cannot be located, no *`PYTHONPATH`* is specified in the environment, and no registry entries can be found, a default path with relative entries is used (e.g. `.\Lib;.\plat-win`, etc).

If a `pyvenv.cfg` file is found alongside the main executable or in the directory one level above the executable, the following variations apply:

- If `home` is an absolute path and *PYTHONHOME* is not set, this path is used instead of the path to the main executable when deducing the home location.

The end result of all this is:

- When running `python.exe`, or any other .exe in the main Python directory (either an installed version, or directly from the PCbuild directory), the core path is deduced, and the core paths in the registry are ignored. Other "application paths" in the registry are always read.
- When Python is hosted in another .exe (different directory, embedded via COM, etc), the "Python Home" will not be deduced, so the core path from the registry is used. Other "application paths" in the registry are always read.
- If Python can't find its home and there are no registry value (frozen .exe, some very strange installation setup) you get a path with some default, but relative, paths.

For those who want to bundle Python into their application or distribution, the following advice will prevent conflicts with other installations:

- Include a `._pth` file alongside your executable containing the directories to include. This will ignore paths listed in the registry and environment variables, and also ignore `site` unless `import site` is listed.
- If you are loading `python3.dll` or `python37.dll` in your own executable, explicitly call `Py_SetPath()` or (at least) `Py_SetProgramName()` before `Py_Initialize()`.
- Clear and/or overwrite *PYTHONPATH* and set *PYTHONHOME* before launching `python.exe` from your application.
- If you cannot use the previous suggestions (for example, you are a distribution that allows people to run `python.exe` directly), ensure that the landmark file (`Lib\os.py`) exists in your install directory. (Note that it will not be detected inside a ZIP file, but a correctly named ZIP file will be detected instead.)

These will ensure that the files in a system-wide installation will not take precedence over the copy of the standard library bundled with your application. Otherwise, your users may experience problems using your application. Note that the first suggestion is the best, as the others may still be susceptible to non-standard paths in the registry and user site-packages.

Changed in version 3.6: Add `._pth` file support and removes `applocal` option from `pyvenv.cfg`.

Changed in version 3.6: Add `python`*XX*`.zip` as a potential landmark when directly adjacent to the executable.

Deprecated since version 3.6: Modules specified in the registry under `Modules` (not `PythonPath`) may be imported by `importlib.machinery.WindowsRegistryFinder`. This finder is enabled on Windows in 3.6.0 and earlier, but may need to be explicitly added to `sys.meta_path` in the future.

4.10 Additional modules

Even though Python aims to be portable among all platforms, there are features that are unique to Windows. A couple of modules, both in the standard library and external, and snippets exist to use these features.

The Windows-specific standard modules are documented in mswin-specific-services.

4.10.1 PyWin32

The PyWin32 module by Mark Hammond is a collection of modules for advanced Windows-specific support. This includes utilities for:

- Component Object Model (COM)
- Win32 API calls
- Registry
- Event log
- Microsoft Foundation Classes (MFC) user interfaces

PythonWin is a sample MFC application shipped with PyWin32. It is an embeddable IDE with a built-in debugger.

See also:

Win32 How Do I…?
by Tim Golden

Python and COM
by David and Paul Boddie

4.10.2 cx_Freeze

cx_Freeze wraps Python scripts into executable Windows programs (`*.exe` files). When you have done this, you can distribute your application without requiring your users to install Python.

4.11 Compiling Python on Windows

If you want to compile CPython yourself, first thing you should do is get the source. You can download either the latest release's source or just grab a fresh checkout.

The source tree contains a build solution and project files for Microsoft Visual Studio, which is the compiler used to build the official Python releases. These files are in the `PCbuild` directory.

Check `PCbuild/readme.txt` for general information on the build process.

For extension modules, consult building-on-windows.

4.12 Other Platforms

With ongoing development of Python, some platforms that used to be supported earlier are no longer supported (due to the lack of users or developers). Check **PEP 11** for details on all unsupported platforms.

- Windows CE is no longer supported since Python 3 (if it ever was).
- The Cygwin installer offers to install the Python interpreter as well

See Python for Windows for detailed information about platforms with pre-compiled installers.

CHAPTER

FIVE

USING PYTHON ON A MAC

Author
Bob Savage <bobsavage@mac.com>

Python on a Mac running macOS is in principle very similar to Python on any other Unix platform, but there are a number of additional features such as the integrated development environment (IDE) and the Package Manager that are worth pointing out.

5.1 Getting and Installing Python

macOS used to come with Python 2.7 pre-installed between versions 10.8 and 12.3. You are invited to install the most recent version of Python 3 from the Python website. A current "universal2 binary" build of Python, which runs natively on the Mac's new Apple Silicon and legacy Intel processors, is available there.

What you get after installing is a number of things:

- A `Python 3.12` folder in your `Applications` folder. In here you find IDLE, the development environment that is a standard part of official Python distributions; and **`Python Launcher`**, which handles double-clicking Python scripts from the Finder.
- A framework `/Library/Frameworks/Python.framework`, which includes the Python executable and libraries. The installer adds this location to your shell path. To uninstall Python, you can remove these three things. A symlink to the Python executable is placed in `/usr/local/bin/`.

Note: On macOS 10.8-12.3, the Apple-provided build of Python is installed in `/System/Library/Frameworks/Python.framework` and `/usr/bin/python`, respectively. You should never modify or delete these, as they are Apple-controlled and are used by Apple- or third-party software. Remember that if you choose to install a newer Python version from python.org, you will have two different but functional Python installations on your computer, so it will be important that your paths and usages are consistent with what you want to do.

IDLE includes a Help menu that allows you to access Python documentation. If you are completely new to Python you should start reading the tutorial introduction in that document.

If you are familiar with Python on other Unix platforms you should read the section on running Python scripts from the Unix shell.

5.1.1 How to run a Python script

Your best way to get started with Python on macOS is through the IDLE integrated development environment; see section *The IDE* and use the Help menu when the IDE is running.

If you want to run Python scripts from the Terminal window command line or from the Finder you first need an editor to create your script. macOS comes with a number of standard Unix command line editors, **vim** **nano** among them. If you want a more Mac-like editor, **BBEdit** from Bare Bones Software (see https://www.barebones.com/products/bbedit/index.html) are good choices, as is **TextMate** (see https://macromates.com). Other editors include **MacVim** (https://macvim.org) and **Aquamacs** (https://aquamacs.org).

To run your script from the Terminal window you must make sure that `/usr/local/bin` is in your shell search path.

To run your script from the Finder you have two options:

- Drag it to **Python Launcher**.
- Select **Python Launcher** as the default application to open your script (or any `.py` script) through the finder Info window and double-click it. **Python Launcher** has various preferences to control how your script is launched. Option-dragging allows you to change these for one invocation, or use its Preferences menu to change things globally.

5.1.2 Running scripts with a GUI

With older versions of Python, there is one macOS quirk that you need to be aware of: programs that talk to the Aqua window manager (in other words, anything that has a GUI) need to be run in a special way. Use **pythonw** instead of **python** to start such scripts.

With Python 3.9, you can use either **python** or **pythonw**.

5.1.3 Configuration

Python on macOS honors all standard Unix environment variables such as *PYTHONPATH*, but setting these variables for programs started from the Finder is non-standard as the Finder does not read your `.profile` or `.cshrc` at startup. You need to create a file `~/.MacOSX/environment.plist`. See Apple's Technical Q&A QA1067 for details.

For more information on installation Python packages, see section *Installing Additional Python Packages*.

5.2 The IDE

Python ships with the standard IDLE development environment. A good introduction to using IDLE can be found at https://www.hashcollision.org/hkn/python/idle_intro/index.html.

5.3 Installing Additional Python Packages

This section has moved to the Python Packaging User Guide.

5.4 GUI Programming

There are several options for building GUI applications on the Mac with Python.

PyObjC is a Python binding to Apple's Objective-C/Cocoa framework, which is the foundation of most modern Mac development. Information on PyObjC is available from pyobjc.

The standard Python GUI toolkit is `tkinter`, based on the cross-platform Tk toolkit (https://www.tcl.tk). An Aqua-native version of Tk is bundled with macOS by Apple, and the latest version can be downloaded and installed from https://www.activestate.com; it can also be built from source.

A number of alternative macOS GUI toolkits are available:

- PySide: Official Python bindings to the Qt GUI toolkit.
- PyQt: Alternative Python bindings to Qt.
- Kivy: A cross-platform GUI toolkit that supports desktop and mobile platforms.
- Toga: Part of the BeeWare Project; supports desktop, mobile, web and console apps.
- wxPython: A cross-platform toolkit that supports desktop operating systems.

5.5 Distributing Python Applications

A range of tools exist for converting your Python code into a standalone distributable application:

- py2app: Supports creating macOS `.app` bundles from a Python project.
- Briefcase: Part of the BeeWare Project; a cross-platform packaging tool that supports creation of `.app` bundles on macOS, as well as managing signing and notarization.
- PyInstaller: A cross-platform packaging tool that creates a single file or folder as a distributable artifact.

5.6 Other Resources

The Pythonmac-SIG mailing list is an excellent support resource for Python users and developers on the Mac:

https://www.python.org/community/sigs/current/pythonmac-sig/

Another useful resource is the MacPython wiki:

https://wiki.python.org/moin/MacPython

CHAPTER

SIX

EDITORS AND IDES

There are a number of IDEs that support Python programming language. Many editors and IDEs provide syntax highlighting, debugging tools, and **PEP 8** checks.

Please go to Python Editors and Integrated Development Environments for a comprehensive list.

APPENDIX

A

GLOSSARY

>>>

The default Python prompt of the interactive shell. Often seen for code examples which can be executed interactively in the interpreter.

...

Can refer to:

- The default Python prompt of the interactive shell when entering the code for an indented code block, when within a pair of matching left and right delimiters (parentheses, square brackets, curly braces or triple quotes), or after specifying a decorator.
- The `Ellipsis` built-in constant.

2to3

A tool that tries to convert Python 2.x code to Python 3.x code by handling most of the incompatibilities which can be detected by parsing the source and traversing the parse tree.

2to3 is available in the standard library as `lib2to3`; a standalone entry point is provided as `Tools/scripts/2to3`. See 2to3-reference.

abstract base class

Abstract base classes complement *duck-typing* by providing a way to define interfaces when other techniques like `hasattr()` would be clumsy or subtly wrong (for example with magic methods). ABCs introduce virtual subclasses, which are classes that don't inherit from a class but are still recognized by `isinstance()` and `issubclass()`; see the `abc` module documentation. Python comes with many built-in ABCs for data structures (in the `collections.abc` module), numbers (in the `numbers` module), streams (in the `io` module), import finders and loaders (in the `importlib.abc` module). You can create your own ABCs with the `abc` module.

annotation

A label associated with a variable, a class attribute or a function parameter or return value, used by convention as a *type hint*.

Annotations of local variables cannot be accessed at runtime, but annotations of global variables, class attributes, and functions are stored in the `__annotations__` special attribute of modules, classes, and functions, respectively.

See *variable annotation*, *function annotation*, **PEP 484** and **PEP 526**, which describe this functionality. Also see annotations-howto for best practices on working with annotations.

argument

A value passed to a *function* (or *method*) when calling the function. There are two kinds of argument:

- *keyword argument*: an argument preceded by an identifier (e.g. `name=`) in a function call or passed as a value in a dictionary preceded by `**`. For example, `3` and `5` are both keyword arguments in the following calls to `complex()`:

```
complex(real=3, imag=5)
complex(**{'real': 3, 'imag': 5})
```

- *positional argument*: an argument that is not a keyword argument. Positional arguments can appear at the beginning of an argument list and/or be passed as elements of an *iterable* preceded by `*`. For example, `3` and `5` are both positional arguments in the following calls:

```
complex(3, 5)
complex(*(3, 5))
```

Arguments are assigned to the named local variables in a function body. See the calls section for the rules governing this assignment. Syntactically, any expression can be used to represent an argument; the evaluated value is assigned to the local variable.

See also the *parameter* glossary entry, the FAQ question on the difference between arguments and parameters, and **PEP 362**.

asynchronous context manager
An object which controls the environment seen in an `async with` statement by defining `__aenter__()` and `__aexit__()` methods. Introduced by **PEP 492**.

asynchronous generator
A function which returns an *asynchronous generator iterator*. It looks like a coroutine function defined with `async def` except that it contains `yield` expressions for producing a series of values usable in an `async for` loop.

Usually refers to an asynchronous generator function, but may refer to an *asynchronous generator iterator* in some contexts. In cases where the intended meaning isn't clear, using the full terms avoids ambiguity.

An asynchronous generator function may contain `await` expressions as well as `async for`, and `async with` statements.

asynchronous generator iterator
An object created by a *asynchronous generator* function.

This is an *asynchronous iterator* which when called using the `__anext__()` method returns an awaitable object which will execute the body of the asynchronous generator function until the next `yield` expression.

Each `yield` temporarily suspends processing, remembering the location execution state (including local variables and pending try-statements). When the *asynchronous generator iterator* effectively resumes with another awaitable returned by `__anext__()`, it picks up where it left off. See **PEP 492** and **PEP 525**.

asynchronous iterable
An object, that can be used in an `async for` statement. Must return an *asynchronous iterator* from its `__aiter__()` method. Introduced by **PEP 492**.

asynchronous iterator
An object that implements the `__aiter__()` and `__anext__()` methods. `__anext__()` must return an *awaitable* object. `async for` resolves the awaitables returned by an asynchronous iterator's `__anext__()` method until it raises a `StopAsyncIteration` exception. Introduced by **PEP 492**.

attribute
A value associated with an object which is usually referenced by name using dotted expressions. For example, if an object *o* has an attribute *a* it would be referenced as *o.a*.

It is possible to give an object an attribute whose name is not an identifier as defined by identifiers, for example using `setattr()`, if the object allows it. Such an attribute will not be accessible using a dotted expression, and would instead need to be retrieved with `getattr()`.

awaitable
An object that can be used in an `await` expression. Can be a *coroutine* or an object with an `__await__()`

method. See also **PEP 492**.

BDFL
Benevolent Dictator For Life, a.k.a. Guido van Rossum, Python's creator.

binary file
A *file object* able to read and write *bytes-like objects*. Examples of binary files are files opened in binary mode (`'rb'`, `'wb'` or `'rb+'`), `sys.stdin.buffer`, `sys.stdout.buffer`, and instances of `io.BytesIO` and `gzip.GzipFile`.

See also *text file* for a file object able to read and write `str` objects.

borrowed reference
In Python's C API, a borrowed reference is a reference to an object, where the code using the object does not own the reference. It becomes a dangling pointer if the object is destroyed. For example, a garbage collection can remove the last *strong reference* to the object and so destroy it.

Calling `Py_INCREF()` on the *borrowed reference* is recommended to convert it to a *strong reference* in-place, except when the object cannot be destroyed before the last usage of the borrowed reference. The `Py_NewRef()` function can be used to create a new *strong reference*.

bytes-like object
An object that supports the bufferobjects and can export a C-*contiguous* buffer. This includes all `bytes`, `bytearray`, and `array.array` objects, as well as many common `memoryview` objects. Bytes-like objects can be used for various operations that work with binary data; these include compression, saving to a binary file, and sending over a socket.

Some operations need the binary data to be mutable. The documentation often refers to these as "read-write bytes-like objects". Example mutable buffer objects include `bytearray` and a `memoryview` of a `bytearray`. Other operations require the binary data to be stored in immutable objects ("read-only bytes-like objects"); examples of these include `bytes` and a `memoryview` of a `bytes` object.

bytecode
Python source code is compiled into bytecode, the internal representation of a Python program in the CPython interpreter. The bytecode is also cached in `.pyc` files so that executing the same file is faster the second time (recompilation from source to bytecode can be avoided). This "intermediate language" is said to run on a *virtual machine* that executes the machine code corresponding to each bytecode. Do note that bytecodes are not expected to work between different Python virtual machines, nor to be stable between Python releases.

A list of bytecode instructions can be found in the documentation for the dis module.

callable
A callable is an object that can be called, possibly with a set of arguments (see *argument*), with the following syntax:

```
callable(argument1, argument2, argumentN)
```

A *function*, and by extension a *method*, is a callable. An instance of a class that implements the `__call__()` method is also a callable.

callback
A subroutine function which is passed as an argument to be executed at some point in the future.

class
A template for creating user-defined objects. Class definitions normally contain method definitions which operate on instances of the class.

class variable
A variable defined in a class and intended to be modified only at class level (i.e., not in an instance of the class).

complex number
An extension of the familiar real number system in which all numbers are expressed as a sum of a real part and an

imaginary part. Imaginary numbers are real multiples of the imaginary unit (the square root of `-1`), often written `i` in mathematics or `j` in engineering. Python has built-in support for complex numbers, which are written with this latter notation; the imaginary part is written with a `j` suffix, e.g., `3+1j`. To get access to complex equivalents of the `math` module, use `cmath`. Use of complex numbers is a fairly advanced mathematical feature. If you're not aware of a need for them, it's almost certain you can safely ignore them.

context manager
An object which controls the environment seen in a `with` statement by defining `__enter__()` and `__exit__()` methods. See **PEP 343**.

context variable
A variable which can have different values depending on its context. This is similar to Thread-Local Storage in which each execution thread may have a different value for a variable. However, with context variables, there may be several contexts in one execution thread and the main usage for context variables is to keep track of variables in concurrent asynchronous tasks. See `contextvars`.

contiguous
A buffer is considered contiguous exactly if it is either *C-contiguous* or *Fortran contiguous*. Zero-dimensional buffers are C and Fortran contiguous. In one-dimensional arrays, the items must be laid out in memory next to each other, in order of increasing indexes starting from zero. In multidimensional C-contiguous arrays, the last index varies the fastest when visiting items in order of memory address. However, in Fortran contiguous arrays, the first index varies the fastest.

coroutine
Coroutines are a more generalized form of subroutines. Subroutines are entered at one point and exited at another point. Coroutines can be entered, exited, and resumed at many different points. They can be implemented with the `async def` statement. See also **PEP 492**.

coroutine function
A function which returns a *coroutine* object. A coroutine function may be defined with the `async def` statement, and may contain `await`, `async for`, and `async with` keywords. These were introduced by **PEP 492**.

CPython
The canonical implementation of the Python programming language, as distributed on python.org. The term "CPython" is used when necessary to distinguish this implementation from others such as Jython or IronPython.

decorator
A function returning another function, usually applied as a function transformation using the `@wrapper` syntax. Common examples for decorators are `classmethod()` and `staticmethod()`.

The decorator syntax is merely syntactic sugar, the following two function definitions are semantically equivalent:

```
def f(arg):
    ...
f = staticmethod(f)

@staticmethod
def f(arg):
    ...
```

The same concept exists for classes, but is less commonly used there. See the documentation for function definitions and class definitions for more about decorators.

descriptor
Any object which defines the methods `__get__()`, `__set__()`, or `__delete__()`. When a class attribute is a descriptor, its special binding behavior is triggered upon attribute lookup. Normally, using *a.b* to get, set or delete an attribute looks up the object named *b* in the class dictionary for *a*, but if *b* is a descriptor, the respective descriptor method gets called. Understanding descriptors is a key to a deep understanding of Python because

they are the basis for many features including functions, methods, properties, class methods, static methods, and reference to super classes.

For more information about descriptors' methods, see descriptors or the Descriptor How To Guide.

dictionary
An associative array, where arbitrary keys are mapped to values. The keys can be any object with `__hash__()` and `__eq__()` methods. Called a hash in Perl.

dictionary comprehension
A compact way to process all or part of the elements in an iterable and return a dictionary with the results. `results = {n: n ** 2 for n in range(10)}` generates a dictionary containing key `n` mapped to value `n ** 2`. See comprehensions.

dictionary view
The objects returned from `dict.keys()`, `dict.values()`, and `dict.items()` are called dictionary views. They provide a dynamic view on the dictionary's entries, which means that when the dictionary changes, the view reflects these changes. To force the dictionary view to become a full list use `list(dictview)`. See dict-views.

docstring
A string literal which appears as the first expression in a class, function or module. While ignored when the suite is executed, it is recognized by the compiler and put into the `__doc__` attribute of the enclosing class, function or module. Since it is available via introspection, it is the canonical place for documentation of the object.

duck-typing
A programming style which does not look at an object's type to determine if it has the right interface; instead, the method or attribute is simply called or used ("If it looks like a duck and quacks like a duck, it must be a duck.") By emphasizing interfaces rather than specific types, well-designed code improves its flexibility by allowing polymorphic substitution. Duck-typing avoids tests using `type()` or `isinstance()`. (Note, however, that duck-typing can be complemented with *abstract base classes*.) Instead, it typically employs `hasattr()` tests or *EAFP* programming.

EAFP
Easier to ask for forgiveness than permission. This common Python coding style assumes the existence of valid keys or attributes and catches exceptions if the assumption proves false. This clean and fast style is characterized by the presence of many `try` and `except` statements. The technique contrasts with the *LBYL* style common to many other languages such as C.

expression
A piece of syntax which can be evaluated to some value. In other words, an expression is an accumulation of expression elements like literals, names, attribute access, operators or function calls which all return a value. In contrast to many other languages, not all language constructs are expressions. There are also *statement*s which cannot be used as expressions, such as `while`. Assignments are also statements, not expressions.

extension module
A module written in C or C++, using Python's C API to interact with the core and with user code.

f-string
String literals prefixed with `'f'` or `'F'` are commonly called "f-strings" which is short for formatted string literals. See also **PEP 498**.

file object
An object exposing a file-oriented API (with methods such as `read()` or `write()`) to an underlying resource. Depending on the way it was created, a file object can mediate access to a real on-disk file or to another type of storage or communication device (for example standard input/output, in-memory buffers, sockets, pipes, etc.). File objects are also called *file-like objects* or *streams*.

There are actually three categories of file objects: raw *binary files*, buffered *binary files* and *text files*. Their interfaces are defined in the `io` module. The canonical way to create a file object is by using the `open()` function.

file-like object
A synonym for *file object*.

filesystem encoding and error handler
Encoding and error handler used by Python to decode bytes from the operating system and encode Unicode to the operating system.

The filesystem encoding must guarantee to successfully decode all bytes below 128. If the file system encoding fails to provide this guarantee, API functions can raise `UnicodeError`.

The `sys.getfilesystemencoding()` and `sys.getfilesystemencodeerrors()` functions can be used to get the filesystem encoding and error handler.

The *filesystem encoding and error handler* are configured at Python startup by the `PyConfig_Read()` function: see `filesystem_encoding` and `filesystem_errors` members of `PyConfig`.

See also the *locale encoding*.

finder
An object that tries to find the *loader* for a module that is being imported.

There are two types of finder: *meta path finders* for use with `sys.meta_path`, and *path entry finders* for use with `sys.path_hooks`.

See importsystem and `importlib` for much more detail.

floor division
Mathematical division that rounds down to nearest integer. The floor division operator is `//`. For example, the expression `11 // 4` evaluates to `2` in contrast to the `2.75` returned by float true division. Note that `(-11) // 4` is `-3` because that is `-2.75` rounded *downward*. See **PEP 238**.

function
A series of statements which returns some value to a caller. It can also be passed zero or more *arguments* which may be used in the execution of the body. See also *parameter*, *method*, and the function section.

function annotation
An *annotation* of a function parameter or return value.

Function annotations are usually used for *type hints*: for example, this function is expected to take two `int` arguments and is also expected to have an `int` return value:

```
def sum_two_numbers(a: int, b: int) -> int:
   return a + b
```

Function annotation syntax is explained in section function.

See *variable annotation* and **PEP 484**, which describe this functionality. Also see annotations-howto for best practices on working with annotations.

__future__
A future statement, `from __future__ import <feature>`, directs the compiler to compile the current module using syntax or semantics that will become standard in a future release of Python. The `__future__` module documents the possible values of *feature*. By importing this module and evaluating its variables, you can see when a new feature was first added to the language and when it will (or did) become the default:

```
>>> import __future__
>>> __future__.division
_Feature((2, 2, 0, 'alpha', 2), (3, 0, 0, 'alpha', 0), 8192)
```

garbage collection
The process of freeing memory when it is not used anymore. Python performs garbage collection via reference

counting and a cyclic garbage collector that is able to detect and break reference cycles. The garbage collector can be controlled using the `gc` module.

generator
A function which returns a *generator iterator*. It looks like a normal function except that it contains `yield` expressions for producing a series of values usable in a for-loop or that can be retrieved one at a time with the `next()` function.

Usually refers to a generator function, but may refer to a *generator iterator* in some contexts. In cases where the intended meaning isn't clear, using the full terms avoids ambiguity.

generator iterator
An object created by a *generator* function.

Each `yield` temporarily suspends processing, remembering the location execution state (including local variables and pending try-statements). When the *generator iterator* resumes, it picks up where it left off (in contrast to functions which start fresh on every invocation).

generator expression
An expression that returns an iterator. It looks like a normal expression followed by a `for` clause defining a loop variable, range, and an optional `if` clause. The combined expression generates values for an enclosing function:

```
>>> sum(i*i for i in range(10))         # sum of squares 0, 1, 4, ... 81
285
```

generic function
A function composed of multiple functions implementing the same operation for different types. Which implementation should be used during a call is determined by the dispatch algorithm.

See also the *single dispatch* glossary entry, the `functools.singledispatch()` decorator, and **PEP 443**.

generic type
A *type* that can be parameterized; typically a container class such as `list` or `dict`. Used for *type hints* and *annotations*.

For more details, see generic alias types, **PEP 483**, **PEP 484**, **PEP 585**, and the `typing` module.

GIL
See *global interpreter lock*.

global interpreter lock
The mechanism used by the *CPython* interpreter to assure that only one thread executes Python *bytecode* at a time. This simplifies the CPython implementation by making the object model (including critical built-in types such as `dict`) implicitly safe against concurrent access. Locking the entire interpreter makes it easier for the interpreter to be multi-threaded, at the expense of much of the parallelism afforded by multi-processor machines.

However, some extension modules, either standard or third-party, are designed so as to release the GIL when doing computationally intensive tasks such as compression or hashing. Also, the GIL is always released when doing I/O.

Past efforts to create a "free-threaded" interpreter (one which locks shared data at a much finer granularity) have not been successful because performance suffered in the common single-processor case. It is believed that overcoming this performance issue would make the implementation much more complicated and therefore costlier to maintain.

hash-based pyc
A bytecode cache file that uses the hash rather than the last-modified time of the corresponding source file to determine its validity. See pyc-invalidation.

hashable
An object is *hashable* if it has a hash value which never changes during its lifetime (it needs a `__hash__()` method), and can be compared to other objects (it needs an `__eq__()` method). Hashable objects which compare equal must have the same hash value.

Hashability makes an object usable as a dictionary key and a set member, because these data structures use the hash value internally.

Most of Python's immutable built-in objects are hashable; mutable containers (such as lists or dictionaries) are not; immutable containers (such as tuples and frozensets) are only hashable if their elements are hashable. Objects which are instances of user-defined classes are hashable by default. They all compare unequal (except with themselves), and their hash value is derived from their `id()`.

IDLE
An Integrated Development and Learning Environment for Python. idle is a basic editor and interpreter environment which ships with the standard distribution of Python.

immutable
An object with a fixed value. Immutable objects include numbers, strings and tuples. Such an object cannot be altered. A new object has to be created if a different value has to be stored. They play an important role in places where a constant hash value is needed, for example as a key in a dictionary.

import path
A list of locations (or *path entries*) that are searched by the *path based finder* for modules to import. During import, this list of locations usually comes from `sys.path`, but for subpackages it may also come from the parent package's `__path__` attribute.

importing
The process by which Python code in one module is made available to Python code in another module.

importer
An object that both finds and loads a module; both a *finder* and *loader* object.

interactive
Python has an interactive interpreter which means you can enter statements and expressions at the interpreter prompt, immediately execute them and see their results. Just launch `python` with no arguments (possibly by selecting it from your computer's main menu). It is a very powerful way to test out new ideas or inspect modules and packages (remember `help(x)`).

interpreted
Python is an interpreted language, as opposed to a compiled one, though the distinction can be blurry because of the presence of the bytecode compiler. This means that source files can be run directly without explicitly creating an executable which is then run. Interpreted languages typically have a shorter development/debug cycle than compiled ones, though their programs generally also run more slowly. See also *interactive*.

interpreter shutdown
When asked to shut down, the Python interpreter enters a special phase where it gradually releases all allocated resources, such as modules and various critical internal structures. It also makes several calls to the *garbage collector*. This can trigger the execution of code in user-defined destructors or weakref callbacks. Code executed during the shutdown phase can encounter various exceptions as the resources it relies on may not function anymore (common examples are library modules or the warnings machinery).

The main reason for interpreter shutdown is that the `__main__` module or the script being run has finished executing.

iterable
An object capable of returning its members one at a time. Examples of iterables include all sequence types (such as `list`, `str`, and `tuple`) and some non-sequence types like `dict`, *file objects*, and objects of any classes you define with an `__iter__()` method or with a `__getitem__()` method that implements *sequence* semantics.

Iterables can be used in a `for` loop and in many other places where a sequence is needed (`zip()`, `map()`, ...). When an iterable object is passed as an argument to the built-in function `iter()`, it returns an iterator for the object. This iterator is good for one pass over the set of values. When using iterables, it is usually not necessary to call `iter()` or deal with iterator objects yourself. The `for` statement does that automatically for you, creating

a temporary unnamed variable to hold the iterator for the duration of the loop. See also *iterator*, *sequence*, and *generator*.

iterator

An object representing a stream of data. Repeated calls to the iterator's `__next__()` method (or passing it to the built-in function `next()`) return successive items in the stream. When no more data are available a `StopIteration` exception is raised instead. At this point, the iterator object is exhausted and any further calls to its `__next__()` method just raise `StopIteration` again. Iterators are required to have an `__iter__()` method that returns the iterator object itself so every iterator is also iterable and may be used in most places where other iterables are accepted. One notable exception is code which attempts multiple iteration passes. A container object (such as a `list`) produces a fresh new iterator each time you pass it to the `iter()` function or use it in a `for` loop. Attempting this with an iterator will just return the same exhausted iterator object used in the previous iteration pass, making it appear like an empty container.

More information can be found in typeiter.

CPython implementation detail: CPython does not consistently apply the requirement that an iterator define `__iter__()`.

key function

A key function or collation function is a callable that returns a value used for sorting or ordering. For example, `locale.strxfrm()` is used to produce a sort key that is aware of locale specific sort conventions.

A number of tools in Python accept key functions to control how elements are ordered or grouped. They include `min()`, `max()`, `sorted()`, `list.sort()`, `heapq.merge()`, `heapq.nsmallest()`, `heapq.nlargest()`, and `itertools.groupby()`.

There are several ways to create a key function. For example. the `str.lower()` method can serve as a key function for case insensitive sorts. Alternatively, a key function can be built from a `lambda` expression such as `lambda r: (r[0], r[2])`. Also, `operator.attrgetter()`, `operator.itemgetter()`, and `operator.methodcaller()` are three key function constructors. See the Sorting HOW TO for examples of how to create and use key functions.

keyword argument

See *argument*.

lambda

An anonymous inline function consisting of a single *expression* which is evaluated when the function is called. The syntax to create a lambda function is `lambda [parameters]: expression`

LBYL

Look before you leap. This coding style explicitly tests for pre-conditions before making calls or lookups. This style contrasts with the *EAFP* approach and is characterized by the presence of many `if` statements.

In a multi-threaded environment, the LBYL approach can risk introducing a race condition between "the looking" and "the leaping". For example, the code, `if key in mapping: return mapping[key]` can fail if another thread removes *key* from *mapping* after the test, but before the lookup. This issue can be solved with locks or by using the EAFP approach.

list

A built-in Python *sequence*. Despite its name it is more akin to an array in other languages than to a linked list since access to elements is $O(1)$.

list comprehension

A compact way to process all or part of the elements in a sequence and return a list with the results. `result = ['{:#04x}'.format(x) for x in range(256) if x % 2 == 0]` generates a list of strings containing even hex numbers (0x..) in the range from 0 to 255. The `if` clause is optional. If omitted, all elements in `range(256)` are processed.

loader
An object that loads a module. It must define a method named `load_module()`. A loader is typically returned by a *finder*. See **PEP 302** for details and `importlib.abc.Loader` for an *abstract base class*.

locale encoding
On Unix, it is the encoding of the LC_CTYPE locale. It can be set with `locale.setlocale(locale.LC_CTYPE, new_locale)`.

On Windows, it is the ANSI code page (ex: `"cp1252"`).

On Android and VxWorks, Python uses `"utf-8"` as the locale encoding.

`locale.getencoding()` can be used to get the locale encoding.

See also the *filesystem encoding and error handler*.

magic method
An informal synonym for *special method*.

mapping
A container object that supports arbitrary key lookups and implements the methods specified in the `collections.abc.Mapping` or `collections.abc.MutableMapping` abstract base classes. Examples include `dict`, `collections.defaultdict`, `collections.OrderedDict` and `collections.Counter`.

meta path finder
A *finder* returned by a search of `sys.meta_path`. Meta path finders are related to, but different from *path entry finders*.

See `importlib.abc.MetaPathFinder` for the methods that meta path finders implement.

metaclass
The class of a class. Class definitions create a class name, a class dictionary, and a list of base classes. The metaclass is responsible for taking those three arguments and creating the class. Most object oriented programming languages provide a default implementation. What makes Python special is that it is possible to create custom metaclasses. Most users never need this tool, but when the need arises, metaclasses can provide powerful, elegant solutions. They have been used for logging attribute access, adding thread-safety, tracking object creation, implementing singletons, and many other tasks.

More information can be found in metaclasses.

method
A function which is defined inside a class body. If called as an attribute of an instance of that class, the method will get the instance object as its first *argument* (which is usually called `self`). See *function* and *nested scope*.

method resolution order
Method Resolution Order is the order in which base classes are searched for a member during lookup. See python_2.3_mro for details of the algorithm used by the Python interpreter since the 2.3 release.

module
An object that serves as an organizational unit of Python code. Modules have a namespace containing arbitrary Python objects. Modules are loaded into Python by the process of *importing*.

See also *package*.

module spec
A namespace containing the import-related information used to load a module. An instance of `importlib.machinery.ModuleSpec`.

MRO
See *method resolution order*.

mutable
Mutable objects can change their value but keep their `id()`. See also *immutable*.

named tuple
The term "named tuple" applies to any type or class that inherits from tuple and whose indexable elements are also accessible using named attributes. The type or class may have other features as well.

Several built-in types are named tuples, including the values returned by `time.localtime()` and `os.stat()`. Another example is `sys.float_info`:

```
>>> sys.float_info[1]                    # indexed access
1024
>>> sys.float_info.max_exp               # named field access
1024
>>> isinstance(sys.float_info, tuple)    # kind of tuple
True
```

Some named tuples are built-in types (such as the above examples). Alternatively, a named tuple can be created from a regular class definition that inherits from `tuple` and that defines named fields. Such a class can be written by hand, or it can be created by inheriting `typing.NamedTuple`, or with the factory function `collections.namedtuple()`. The latter techniques also add some extra methods that may not be found in hand-written or built-in named tuples.

namespace
The place where a variable is stored. Namespaces are implemented as dictionaries. There are the local, global and built-in namespaces as well as nested namespaces in objects (in methods). Namespaces support modularity by preventing naming conflicts. For instance, the functions `builtins.open` and `os.open()` are distinguished by their namespaces. Namespaces also aid readability and maintainability by making it clear which module implements a function. For instance, writing `random.seed()` or `itertools.islice()` makes it clear that those functions are implemented by the `random` and `itertools` modules, respectively.

namespace package
A **PEP 420** *package* which serves only as a container for subpackages. Namespace packages may have no physical representation, and specifically are not like a *regular package* because they have no `__init__.py` file.

See also *module*.

nested scope
The ability to refer to a variable in an enclosing definition. For instance, a function defined inside another function can refer to variables in the outer function. Note that nested scopes by default work only for reference and not for assignment. Local variables both read and write in the innermost scope. Likewise, global variables read and write to the global namespace. The `nonlocal` allows writing to outer scopes.

new-style class
Old name for the flavor of classes now used for all class objects. In earlier Python versions, only new-style classes could use Python's newer, versatile features like `__slots__`, descriptors, properties, `__getattribute__()`, class methods, and static methods.

object
Any data with state (attributes or value) and defined behavior (methods). Also the ultimate base class of any *new-style class*.

package
A Python *module* which can contain submodules or recursively, subpackages. Technically, a package is a Python module with a `__path__` attribute.

See also *regular package* and *namespace package*.

parameter
A named entity in a *function* (or method) definition that specifies an *argument* (or in some cases, arguments) that

the function can accept. There are five kinds of parameter:

- *positional-or-keyword*: specifies an argument that can be passed either *positionally* or as a *keyword argument*. This is the default kind of parameter, for example *foo* and *bar* in the following:

```
def func(foo, bar=None): ...
```

- *positional-only*: specifies an argument that can be supplied only by position. Positional-only parameters can be defined by including a `/` character in the parameter list of the function definition after them, for example *posonly1* and *posonly2* in the following:

```
def func(posonly1, posonly2, /, positional_or_keyword): ...
```

- *keyword-only*: specifies an argument that can be supplied only by keyword. Keyword-only parameters can be defined by including a single var-positional parameter or bare `*` in the parameter list of the function definition before them, for example *kw_only1* and *kw_only2* in the following:

```
def func(arg, *, kw_only1, kw_only2): ...
```

- *var-positional*: specifies that an arbitrary sequence of positional arguments can be provided (in addition to any positional arguments already accepted by other parameters). Such a parameter can be defined by prepending the parameter name with `*`, for example *args* in the following:

```
def func(*args, **kwargs): ...
```

- *var-keyword*: specifies that arbitrarily many keyword arguments can be provided (in addition to any keyword arguments already accepted by other parameters). Such a parameter can be defined by prepending the parameter name with `**`, for example *kwargs* in the example above.

Parameters can specify both optional and required arguments, as well as default values for some optional arguments.

See also the *argument* glossary entry, the FAQ question on the difference between arguments and parameters, the `inspect.Parameter` class, the function section, and **PEP 362**.

path entry
A single location on the *import path* which the *path based finder* consults to find modules for importing.

path entry finder
A *finder* returned by a callable on `sys.path_hooks` (i.e. a *path entry hook*) which knows how to locate modules given a *path entry*.

See `importlib.abc.PathEntryFinder` for the methods that path entry finders implement.

path entry hook
A callable on the `sys.path_hooks` list which returns a *path entry finder* if it knows how to find modules on a specific *path entry*.

path based finder
One of the default *meta path finders* which searches an *import path* for modules.

path-like object
An object representing a file system path. A path-like object is either a `str` or `bytes` object representing a path, or an object implementing the `os.PathLike` protocol. An object that supports the `os.PathLike` protocol can be converted to a `str` or `bytes` file system path by calling the `os.fspath()` function; `os.fsdecode()` and `os.fsencode()` can be used to guarantee a `str` or `bytes` result instead, respectively. Introduced by **PEP 519**.

PEP
Python Enhancement Proposal. A PEP is a design document providing information to the Python community,

or describing a new feature for Python or its processes or environment. PEPs should provide a concise technical specification and a rationale for proposed features.

PEPs are intended to be the primary mechanisms for proposing major new features, for collecting community input on an issue, and for documenting the design decisions that have gone into Python. The PEP author is responsible for building consensus within the community and documenting dissenting opinions.

See **PEP 1**.

portion
A set of files in a single directory (possibly stored in a zip file) that contribute to a namespace package, as defined in **PEP 420**.

positional argument
See *argument*.

provisional API
A provisional API is one which has been deliberately excluded from the standard library's backwards compatibility guarantees. While major changes to such interfaces are not expected, as long as they are marked provisional, backwards incompatible changes (up to and including removal of the interface) may occur if deemed necessary by core developers. Such changes will not be made gratuitously – they will occur only if serious fundamental flaws are uncovered that were missed prior to the inclusion of the API.

Even for provisional APIs, backwards incompatible changes are seen as a "solution of last resort" - every attempt will still be made to find a backwards compatible resolution to any identified problems.

This process allows the standard library to continue to evolve over time, without locking in problematic design errors for extended periods of time. See **PEP 411** for more details.

provisional package
See *provisional API*.

Python 3000
Nickname for the Python 3.x release line (coined long ago when the release of version 3 was something in the distant future.) This is also abbreviated "Py3k".

Pythonic
An idea or piece of code which closely follows the most common idioms of the Python language, rather than implementing code using concepts common to other languages. For example, a common idiom in Python is to loop over all elements of an iterable using a `for` statement. Many other languages don't have this type of construct, so people unfamiliar with Python sometimes use a numerical counter instead:

```
for i in range(len(food)):
    print(food[i])
```

As opposed to the cleaner, Pythonic method:

```
for piece in food:
    print(piece)
```

qualified name
A dotted name showing the "path" from a module's global scope to a class, function or method defined in that module, as defined in **PEP 3155**. For top-level functions and classes, the qualified name is the same as the object's name:

```
>>> class C:
...     class D:
...         def meth(self):
...             pass
```

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```
...
>>> C.__qualname__
'C'
>>> C.D.__qualname__
'C.D'
>>> C.D.meth.__qualname__
'C.D.meth'
```

When used to refer to modules, the *fully qualified name* means the entire dotted path to the module, including any parent packages, e.g. `email.mime.text`:

```
>>> import email.mime.text
>>> email.mime.text.__name__
'email.mime.text'
```

reference count

The number of references to an object. When the reference count of an object drops to zero, it is deallocated. Some objects are "immortal" and have reference counts that are never modified, and therefore the objects are never deallocated. Reference counting is generally not visible to Python code, but it is a key element of the *CPython* implementation. Programmers can call the `sys.getrefcount()` function to return the reference count for a particular object.

regular package

A traditional *package*, such as a directory containing an `__init__.py` file.

See also *namespace package*.

__slots__

A declaration inside a class that saves memory by pre-declaring space for instance attributes and eliminating instance dictionaries. Though popular, the technique is somewhat tricky to get right and is best reserved for rare cases where there are large numbers of instances in a memory-critical application.

sequence

An *iterable* which supports efficient element access using integer indices via the `__getitem__()` special method and defines a `__len__()` method that returns the length of the sequence. Some built-in sequence types are `list`, `str`, `tuple`, and `bytes`. Note that `dict` also supports `__getitem__()` and `__len__()`, but is considered a mapping rather than a sequence because the lookups use arbitrary *immutable* keys rather than integers.

The `collections.abc.Sequence` abstract base class defines a much richer interface that goes beyond just `__getitem__()` and `__len__()`, adding `count()`, `index()`, `__contains__()`, and `__reversed__()`. Types that implement this expanded interface can be registered explicitly using `register()`. For more documentation on sequence methods generally, see Common Sequence Operations.

set comprehension

A compact way to process all or part of the elements in an iterable and return a set with the results. `results = {c for c in 'abracadabra' if c not in 'abc'}` generates the set of strings `{'r', 'd'}`. See comprehensions.

single dispatch

A form of *generic function* dispatch where the implementation is chosen based on the type of a single argument.

slice

An object usually containing a portion of a *sequence*. A slice is created using the subscript notation, `[]` with colons between numbers when several are given, such as in `variable_name[1:3:5]`. The bracket (subscript) notation uses `slice` objects internally.

special method

A method that is called implicitly by Python to execute a certain operation on a type, such as addition. Such methods

have names starting and ending with double underscores. Special methods are documented in specialnames.

statement
A statement is part of a suite (a "block" of code). A statement is either an *expression* or one of several constructs with a keyword, such as `if`, `while` or `for`.

static type checker
An external tool that reads Python code and analyzes it, looking for issues such as incorrect types. See also *type hints* and the `typing` module.

strong reference
In Python's C API, a strong reference is a reference to an object which is owned by the code holding the reference. The strong reference is taken by calling `Py_INCREF()` when the reference is created and released with `Py_DECREF()` when the reference is deleted.

The `Py_NewRef()` function can be used to create a strong reference to an object. Usually, the `Py_DECREF()` function must be called on the strong reference before exiting the scope of the strong reference, to avoid leaking one reference.

See also *borrowed reference*.

text encoding
A string in Python is a sequence of Unicode code points (in range `U+0000`–`U+10FFFF`). To store or transfer a string, it needs to be serialized as a sequence of bytes.

Serializing a string into a sequence of bytes is known as "encoding", and recreating the string from the sequence of bytes is known as "decoding".

There are a variety of different text serialization codecs, which are collectively referred to as "text encodings".

text file
A *file object* able to read and write `str` objects. Often, a text file actually accesses a byte-oriented datastream and handles the *text encoding* automatically. Examples of text files are files opened in text mode (`'r'` or `'w'`), `sys.stdin`, `sys.stdout`, and instances of `io.StringIO`.

See also *binary file* for a file object able to read and write *bytes-like objects*.

triple-quoted string
A string which is bound by three instances of either a quotation mark (") or an apostrophe ('). While they don't provide any functionality not available with single-quoted strings, they are useful for a number of reasons. They allow you to include unescaped single and double quotes within a string and they can span multiple lines without the use of the continuation character, making them especially useful when writing docstrings.

type
The type of a Python object determines what kind of object it is; every object has a type. An object's type is accessible as its `__class__` attribute or can be retrieved with `type(obj)`.

type alias
A synonym for a type, created by assigning the type to an identifier.

Type aliases are useful for simplifying *type hints*. For example:

```
def remove_gray_shades(
        colors: list[tuple[int, int, int]]) -> list[tuple[int, int, int]]:
    pass
```

could be made more readable like this:

```
Color = tuple[int, int, int]
```

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```
def remove_gray_shades(colors: list[Color]) -> list[Color]:
    pass
```

See `typing` and **PEP 484**, which describe this functionality.

type hint

An *annotation* that specifies the expected type for a variable, a class attribute, or a function parameter or return value.

Type hints are optional and are not enforced by Python but they are useful to *static type checkers*. They can also aid IDEs with code completion and refactoring.

Type hints of global variables, class attributes, and functions, but not local variables, can be accessed using `typing.get_type_hints()`.

See `typing` and **PEP 484**, which describe this functionality.

universal newlines

A manner of interpreting text streams in which all of the following are recognized as ending a line: the Unix end-of-line convention `'\n'`, the Windows convention `'\r\n'`, and the old Macintosh convention `'\r'`. See **PEP 278** and **PEP 3116**, as well as `bytes.splitlines()` for an additional use.

variable annotation

An *annotation* of a variable or a class attribute.

When annotating a variable or a class attribute, assignment is optional:

```
class C:
    field: 'annotation'
```

Variable annotations are usually used for *type hints*: for example this variable is expected to take `int` values:

```
count: int = 0
```

Variable annotation syntax is explained in section annassign.

See *function annotation*, **PEP 484** and **PEP 526**, which describe this functionality. Also see annotations-howto for best practices on working with annotations.

virtual environment

A cooperatively isolated runtime environment that allows Python users and applications to install and upgrade Python distribution packages without interfering with the behaviour of other Python applications running on the same system.

See also `venv`.

virtual machine

A computer defined entirely in software. Python's virtual machine executes the *bytecode* emitted by the bytecode compiler.

Zen of Python

Listing of Python design principles and philosophies that are helpful in understanding and using the language. The listing can be found by typing "`import this`" at the interactive prompt.

APPENDIX

B

ABOUT THESE DOCUMENTS

These documents are generated from reStructuredText sources by Sphinx, a document processor specifically written for the Python documentation.

Development of the documentation and its toolchain is an entirely volunteer effort, just like Python itself. If you want to contribute, please take a look at the reporting-bugs page for information on how to do so. New volunteers are always welcome!

Many thanks go to:

- Fred L. Drake, Jr., the creator of the original Python documentation toolset and writer of much of the content;
- the Docutils project for creating reStructuredText and the Docutils suite;
- Fredrik Lundh for his Alternative Python Reference project from which Sphinx got many good ideas.

B.1 Contributors to the Python Documentation

Many people have contributed to the Python language, the Python standard library, and the Python documentation. See Misc/ACKS in the Python source distribution for a partial list of contributors.

It is only with the input and contributions of the Python community that Python has such wonderful documentation – Thank You!

APPENDIX

C

HISTORY AND LICENSE

C.1 History of the software

Python was created in the early 1990s by Guido van Rossum at Stichting Mathematisch Centrum (CWI, see https://www.cwi.nl/) in the Netherlands as a successor of a language called ABC. Guido remains Python's principal author, although it includes many contributions from others.

In 1995, Guido continued his work on Python at the Corporation for National Research Initiatives (CNRI, see https://www.cnri.reston.va.us/) in Reston, Virginia where he released several versions of the software.

In May 2000, Guido and the Python core development team moved to BeOpen.com to form the BeOpen PythonLabs team. In October of the same year, the PythonLabs team moved to Digital Creations (now Zope Corporation; see https://www.zope.org/). In 2001, the Python Software Foundation (PSF, see https://www.python.org/psf/) was formed, a non-profit organization created specifically to own Python-related Intellectual Property. Zope Corporation is a sponsoring member of the PSF.

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2.1.2	2.1.1	2002	PSF	yes
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C.3.1 Mersenne Twister

The `_random` C extension underlying the `random` module includes code based on a download from http://www.math.sci.hiroshima-u.ac.jp/~m-mat/MT/MT2002/emt19937ar.html. The following are the verbatim comments from the original code:

```
A C-program for MT19937, with initialization improved 2002/1/26.
Coded by Takuji Nishimura and Makoto Matsumoto.

Before using, initialize the state by using init_genrand(seed)
or init_by_array(init_key, key_length).

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```

C.3.2 Sockets

The `socket` module uses the functions, `getaddrinfo()`, and `getnameinfo()`, which are coded in separate source files from the WIDE Project, https://www.wide.ad.jp/.

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- Arguments more compliant with Python standard
```

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C.3.9 Select kqueue

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C.3.10 SipHash24

The file `Python/pyhash.c` contains Marek Majkowski' implementation of Dan Bernstein's SipHash24 algorithm. It contains the following note:

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Original location:
   https://github.com/majek/csiphash/

Solution inspired by code from:
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   djb (supercop/crypto_auth/siphash24/little2)
   Jean-Philippe Aumasson (https://131002.net/siphash/siphash24.c)
```

C.3.11 strtod and dtoa

The file `Python/dtoa.c`, which supplies C functions dtoa and strtod for conversion of C doubles to and from strings, is derived from the file of the same name by David M. Gay, currently available from https://web.archive.org/web/20220517033456/http://www.netlib.org/fp/dtoa.c. The original file, as retrieved on March 16, 2009, contains the following copyright and licensing notice:

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```
 *
 ************************************************************/
```

C.3.12 OpenSSL

The modules `hashlib`, `posix`, `ssl`, `crypt` use the OpenSSL library for added performance if made available by the operating system. Additionally, the Windows and macOS installers for Python may include a copy of the OpenSSL libraries, so we include a copy of the OpenSSL license here. For the OpenSSL 3.0 release, and later releases derived from that, the Apache License v2 applies:

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C.3.16 cfuhash

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C.3.17 libmpdec

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C.3.18 W3C C14N test suite

The C14N 2.0 test suite in the `test` package (`Lib/test/xmltestdata/c14n-20/`) was retrieved from the W3C website at https://www.w3.org/TR/xml-c14n2-testcases/ and is distributed under the 3-clause BSD license:

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```

C.3.19 Audioop

The audioop module uses the code base in g771.c file of the SoX project. https://sourceforge.net/projects/sox/files/sox/12.17.7/sox-12.17.7.tar.gz

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C.3.20 asyncio

Parts of the `asyncio` module are incorporated from uvloop 0.16, which is distributed under the MIT license:

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